

Development of a Small-Scale Underwater Remotely Operated Vehicle (ROV)

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CONTENTS

I	Introduction	1
I-A	Context & Motivation:	1
I-B	Problem Statement:	1
I-C	Objectives & Scope:	2
I-D	Team Dynamics:	2
II	Literature Review	2
II-A	Literature Review Summary	2
II-B	Evaluation of Literature Review Findings and New Literature	3
III	Methodology & System Design	4
III-A	Design Goals and Iterative Design Process	4
III-B	Software and Hardware Selection	5
III-B1	Design and Simulation	5
III-B2	Control Software	5
III-C	Material Selection	5
III-D	Electronic System Selection and Design	6
III-D1	Payload	6
III-D2	Power Distribution and Safety Control Loop	6
III-D3	Flight Control and Propulsion	6
III-D4	Payload and Video Integration	6
III-D5	Communication Architecture	7
III-D6	System Architecture	7
IV	Implementation & Prototyping	7
IV-A	Design Phase	7
IV-B	Production Phase	7
IV-B1	Development Roadblocks	7
IV-B2	Buoyancy and Stability Design	9
IV-C	Design Specification	11
V	Testing & Results	11
V-1	Thruster, Payload and Kill Switch Testing	11
V-2	Simulation Testing	12
V-3	Physical Testing	14
V-4	Testing Comparison	15
VI	Discussion and Project Evaluation	16
VI-1	Safety and Risks	16
VI-2	Environmental Impact	16
VI-3	Limitations and Weaknesses	16
VI-4	Goals and Objectives	18
VII	Conclusion	18
	References	19
VIII	Appendices	20
VIII-A	Appendix A: Original Literature Review (from Project Proposal)	20
VIII-A1	Introduction	20
VIII-A2	Methodology and Scope	21
VIII-A3	Key Design Themes	21
VIII-A4	Hardware Selections	22
VIII-A5	Software and Communication Frameworks	22
VIII-A6	Control System Architectures	23
VIII-A7	System Modelling Approaches	23
VIII-B	Appendix B: Comprehensive Bill of Materials (BOM) & Full Budget	23
VIII-C	Appendix C: Project Timelines	25
VIII-D	Appendix D: CAD Models	27
VIII-E	Appendix E: Risk Assessment	29

Development of a Small-Scale Underwater ROV for Research Purposes

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Abstract—This paper presents the development of the SubbyROV, a small-scale, cost-effective Remotely Operated Vehicle (ROV) designed to address the logistical and financial limitations of utilizing large-scale vehicles for academic research and educational outreach at the Queensland University of Technology (QUT). While existing platforms like the BlueROV2 are highly capable, their size often impedes rapid prototyping and algorithm testing due to deployment complexities. To bridge this gap, an iterative systems engineering approach was employed to design a highly modular, miniature ROV leveraging Commercial Off-The-Shelf (COTS) components and custom 3D-printed Polylactic Acid (PLA) structures. The mechanical design features a compact 3-inch acrylic pressure hull housing densely integrated electrical components, including a Pixhawk flight controller, Fathom-X tether interfaces, and an internal high-capacity battery. A significant innovation was the adoption of an underactuated four-thruster configuration, which drastically minimized the vehicle's footprint while retaining essential maneuverability. The software architecture utilizes ArduSub and QGroundControl for teleoperation, supported by extensive hydrodynamic modeling and a custom Stonefish digital twin to validate buoyancy and dynamic stability prior to physical deployment. Physical testing in a controlled pool environment confirmed the waterproof integrity of the pressure hull and successfully demonstrated the vehicle's maneuverability under manual control. The results validate the SubbyROV as a viable, low-cost testbed. Ultimately, this project establishes a robust hardware and simulation foundation, setting the stage for future research into non-linear underactuated control systems, autonomous navigation, and machine learning integration.

Index Terms—Remotely Operated Vehicle (ROV), Systems Engineering, QGroundControl, ArduSub, Underactuated Control, Underwater Robotics, Small Scale.

I. INTRODUCTION

A. Context & Motivation:

Unmanned Underwater Vehicles (UUVs) and Remotely Operated Vehicles (ROVs) have become indispensable tools for a vast range of applications, including oceanographic surveys, infrastructure maintenance, and defense operations. As technology has advanced, these vehicles have evolved from simple teleoperated platforms into complex autonomous systems capable of executing sophisticated missions. However, much of the research and commercial development has focused on large-scale, work-class vehicles or highly specialized Autonomous Underwater Vehicles (AUVs). At the Queensland University of Technology (QUT), the existing BlueROV2 platform, while highly capable, presents several limitations for rapid prototyping and educational outreach. Its large size makes it cumbersome to deploy, and any testing requires formal pool approval and significant logistical overhead. For small-scale experiments, algorithm testing, and classroom demonstrations, the BlueROV2 is often an overkill solution that impedes the pace of research.

B. Problem Statement:

Consequently, there is a distinct gap in the availability of small-scale, low-cost, and modular ROV platforms that are computationally equipped for modern research. To address this, there is a clear need to design and develop a miniature, modular ROV platform constructed primarily from Commercial Off-The-Shelf (COTS) components. This platform must maintain the agility and sensor capabilities required to act as a viable testbed

for advanced control and perception algorithms, while being small enough to be easily deployed in confined testing tanks without the extensive logistical requirements of larger vehicles.

C. Objectives & Scope:

The primary objective of this project is to create a fully operational, small-scale underwater ROV to serve as both a demonstrator and a flexible research testbed at QUT. Key objectives include:

- **Controlled Maneuverability:** Achieving stable, controlled diving and movement using a thruster configuration.
- **Autonomous Capabilities:** Enabling the execution of pre-programmed missions and providing the necessary hardware overhead to support future autonomy pipelines.
- **Modularity and Accessibility:** Ensuring the design is easily modifiable, utilizing open-source frameworks such as ROS2, ArduSub/PX4, and MAVLink for seamless integration of future sensors and payloads.
- **Demonstrator Platform:** Serving as an educational tool for QUT demonstrations to showcase underwater robotics principles in a highly portable format.
- **Document Finding:** The findings from this project will be open sourced, including the 3D SolidWorks models, ROS2 simulation files and all documentation and findings.

An original and revised project timeline found in VIII-B outlines the major tasks to be completed and their estimated duration. These timelines were used when discussing with stakeholders and making decisions about requesting support and refining scope.

D. Team Dynamics:

As the project transitioned from the initial design phase in the first semester to prototyping and system integration in the second semester, the scope and complexity of the development necessitated a more collaborative systems engineering approach. A key design challenge was ensuring all electronic components could fit inside the pressure housing. The resolution of this challenge required additional time that was not accounted for in the project timeline. This led to an additional team

member being requested to help in the design of the digital twin and simulation software programming to allow the team to meet the deliverables for the project. This evolution allowed for a more robust division of labor, enabling parallel development of the mechanical chassis, electrical integration, and digital simulation capabilities to meet the project's ambitious deliverables.

II. LITERATURE REVIEW

A. Literature Review Summary

A comprehensive survey of existing literature reveals a distinct gap in the field of underwater robotics for small-scale, low-cost, and modular Remotely Operated Vehicles (ROVs) that are also computationally equipped for modern research. While the domain is populated with large, work-class vehicles for industrial applications and highly specialised Autonomous Underwater Vehicles (AUVs), these platforms are often financially and logistically prohibitive for academic research and rapid prototyping [1]. This review synthesised key design philosophies and technical trade-offs from a range of academic papers and technical reports, establishing a theoretical foundation to inform the design of a novel research platform.

A primary theme is the fundamental decision in chassis architecture, which presents a trade-off between modularity and hydrodynamic performance. Open-frame designs are consistently favoured for research and educational platforms due to their inherent modularity, which simplifies the process of mounting, testing, and reconfiguring sensors, thrusters, and other payloads, as seen in vehicles like the eROV [2]. This adaptability is paramount for a vehicle intended to serve as a testbed. Conversely, enclosed, torpedo-style hulls are chosen when operational efficiency, speed, and endurance are the primary drivers, as their streamlined form minimises drag [3], [4]. For this project's objectives, the principles of the open-frame design are more closely aligned with the need for a flexible and extensible research platform.

Maneuverability in a compact form factor is another critical consideration, heavily influenced by thruster configuration. While simple 3-thruster systems can provide basic motion [5], the literature indicates that full 6-DOF control is a key

enabler for complex inspection and intervention tasks. This is typically achieved with six or more thrusters [6], [7]. A common and effective strategy is the use of a vectored thruster arrangement, where horizontal thrusters are angled to provide coupled control of surge, sway, and yaw [1], [8]. This approach allows for high agility and omnidirectional movement without a large physical footprint, a key idea that heavily influenced this project's propulsion system design.

The selection of onboard hardware reflects a clear scalability based on mission complexity and budget. At the lower end, simple microcontrollers like Arduino are sufficient for basic teleoperation and sensor reading [5], [9]. However, to serve as a viable testbed for modern algorithms, particularly in computer vision and machine learning (ML), a more powerful single-board computer, such as a Raspberry Pi or Nvidia Jetson, is essential. These platforms provide the necessary processing power for real-time video streaming, network communication, and running inference models [8], [10]–[12]. This distinction highlights the necessity of selecting a companion computer that not only meets current needs but also provides headroom for future computational expansion.

Finally, the software architecture and control system design are integral to the vehicle's capability. The Robot Operating System (ROS) is frequently cited as the framework of choice for complex, modular robotic systems due to its robust inter-process communication, extensive libraries, and available hardware drivers [7], [11]. While simpler platforms may use custom firmware, open-source solutions like ArduSub offer a powerful, community-supported alternative that can significantly accelerate development [13]. At the control level, the Proportional-Integral-Derivative (PID) controller is a foundational and ubiquitous method for achieving basic stability, such as depth and heading hold [6], [10]. However, its performance can degrade in the presence of external disturbances, leading more advanced projects to explore robust control techniques like Sliding Mode Control or Fuzzy Logic controllers [5], [14]. These established ideas from the literature provide a clear benchmark against which this project's design choices and practical progress can be evaluated. The full literature review is available in Appendix

VIII-A for further detail.

B. Evaluation of Literature Review Findings and New Literature

During the first phase of the project, it was found that the insights from the original literature review were largely applicable to the design and development of the medium scale ROV but not small scale ROVs. Using 6 thrusters to achieve 6 DOF was found to be impractical for a small scale ROV due to size and power consumption constraints. This was also true for the use of a companion computer such as a Raspberry Pi or Nvidia Jetson. These components, while useful for ML applications, were found to be too large and power hungry for the size of ROV being designed. This led to further literature review to find alternatives that would suit the size and power constraints of the small scale ROV.

A paper by Sujan et al. (2024) [15] presented a design for a small scale ROV that used 4 thrusters in a vectored configuration to achieve underactuated control of the ROV as seen in Figure 1. This design was found to be more suitable for the project as it allowed for a smaller footprint and lower power consumption, however the underactuation resulted in the lack of independent pitch control.

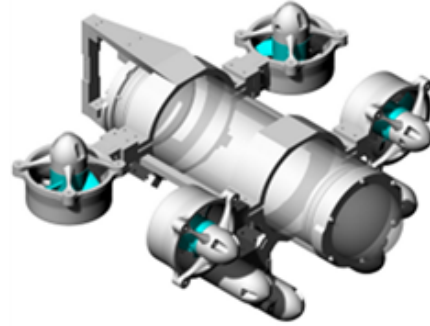


Fig. 1. 4-thruster vectored configuration for a small-scale ROV, demonstrating an underactuated layout to minimise footprint and power consumption. Adapted from Sujan et al. (2024) [15].

An extension of this approach was considered by using 3D printed 45-degree brackets to provide each thruster with a vectored force. The benefit of this design is that it reduces the number of thrusters from six to four, thereby reducing the

size of the battery and the required electronic speed controllers (ESCs). This is a key area of improvement for the small-scale ROV, as it allows for a smaller hull, a reduced overall size, and cost reductions. However this approach was not backed by current literature and was eventually discarded in favour of a traditional 4-thruster configuration.

Due to sizing constraints found during the initial design phase, lithium-ion battery packs were found to be too large and had to be replaced with a smaller Lithium-ion Polymer (LiPo) battery. While LiPo batteries were selected for their high energy density and discharge rates, their use introduces significant safety considerations. An issues paper by the Australian Competition & Consumer Commission (ACCC) highlights the primary hazard associated with Li-ion batteries as 'thermal runaway', a rapid, self-sustaining temperature increase [16]. This event can be triggered by physical impact (a key risk during ROV operations) or water ingress leading to a short circuit. For this project, these risks underscore the criticality of a robust, impact-resistant, and completely watertight enclosure for the battery, as well as strict adherence to balanced charging protocols to mitigate these hazards and ensure the safe operation of the ROV.

Additionally, new literature on the MAVLink communication protocol was found, as the small size ROV was not large enough to house a full companion computer. This led to the use of a Pixhawk flight controller as the main computer, which utilizes the MAVLink protocol to communicate with the surface station. MAVLink is a lightweight, header-only message marshalling library initially designed for micro air vehicles [17]. This protocol is widely used in the drone community and has been adapted for use in underwater vehicles. The use of MAVLink allows for a reliable and efficient communication link between the ROV and the surface station, which is essential for remote operation and data transfer.

Furthermore, additional literature was reviewed in the context of recent project developments:

Ethernet Integration: In miniature ROV architectures where spatial constraints prevent large umbilical tethers, internal networking must be strictly managed. Recent developments emphasize the use of compact Ethernet switches or splitters

to multiplex communication channels between the surface control station, the companion computer, and high-bandwidth payloads such as IP cameras or imaging sonars over a single tether [18].

Simulation Environments: To validate control systems prior to physical deployment, highly accurate simulation environments are required. The Stonefish simulator has emerged as a state-of-the-art tool specifically designed for marine robotics, offering realistic hydrodynamics, sensor simulations, and native ROS/ROS2 interfaces [19]. Recent updates to Stonefish have further expanded its capabilities to support machine learning research, improved thruster modeling, and advanced tether dynamics, making it an ideal platform for testing small-scale underwater vehicles in complex environments [20].

Underactuated Control: Reducing the number of thrusters to save weight and power necessitates the use of underactuated control methods. Such methods must account for the highly nonlinear and coupled dynamics of underwater vehicles operating with fewer control inputs than degrees of freedom. Recent studies have demonstrated the efficacy of nonlinear model predictive control (MPC) and behavior trees for achieving complex, agile maneuvers in underactuated AUVs [21], [22]. Additionally, finite-time tracking control techniques utilizing command filters and disturbance observers have been proposed to mitigate the effects of unmodeled dynamics and ocean currents [23]. Sliding mode controllers have also been validated for managing heading and diving planes in underactuated systems, ensuring robust performance under uncertain conditions [24].

III. METHODOLOGY & SYSTEM DESIGN

The methodology employed in this project followed the principles of systems engineering, with a strong emphasis on literature review, iterative design, and rigorous testing.

A. Design Goals and Iterative Design Process

The development of the ROV adhered to a systems engineering design framework centered around two primary mechanisms for progression: iterative design and stakeholder approval. The philosophy of iterative design recognizes that com-

plex engineering systems, particularly those operating in challenging environments like underwater, rarely achieve optimal configuration in a single pass. By employing continuous cycles of modeling, feedback, and refinement, potential integration issues can be identified early and corrected systematically. To complement this, stakeholder approval served as the primary quality and requirements gate. Rather than relying solely on internal verification, formal reviews with key stakeholders ensured that design decisions remained closely aligned with the overarching research goals, budgetary constraints, and operational safety requirements before the project could transition into physical fabrication and development. Specifically, for each cycle a 3D model supporting the design was produced and a meeting with the supervisor was held to discuss the design. If the design passed it would move into development and if changes needed to be made to meet the stakeholders' ideal criteria these were documented in detail.

B. Software and Hardware Selection

1) *Design and Simulation:* For this project, intensive 3D modeling was required; SolidWorks 2025 was selected as the main modeling tool. This choice was based on several factors, with the main being its mass properties function, which can dynamically calculate the center of mass (COM) and, with some additional techniques, the center of buoyancy (COB). To support this software, Blender was also used to export the 3D model into a watertight 3D mesh for dynamic simulation tools. For digital twin simulation, the open-source software used was Stonefish [19], specifically the ROS2 interface of Stonefish. Once the weight parameters of the design were calculated with SolidWorks, the 3D model was exported and imported into Blender, where any holes in the mesh were filled using Blender's remesh tools. This STL file was then imported into a customized scene programmed in the Stonefish package, with its sensors and thruster locations being added to the STL model. The main strength of this method is that Stonefish calculates the hydrodynamics based on the STL model and the weight parameters, completely removing the need for manual hydrodynamic calculations or expensive and time-consuming CFD simulations. While this means

additional work is required to evaluate all the necessary dampening parameters for the 6-DOF modeling approach, this was deemed an acceptable trade-off for reducing the time required to achieve accurate simulations.

2) *Control Software:* This project followed similar control software structures to literature, utilizing QGroundControl as the main ground control station software for operation and manual control. This open-source software, commonly used for robotic and drone control, was chosen due to its extensive support for the Pixhawk flight controller architecture. This was paired with ArduSub, an open-source firmware built on top of the ArduPilot project, which provides the necessary control algorithms and interfaces for ROV operation. QGroundControl was used to load the ArduSub software onto the Pixhawk flight controller via USB. Once connected, the Pixhawk's parameters were modified via QGroundControl. Another strength was connecting the camera stream to QGroundControl, with plans to also control a 3-axis camera gimbal using commands sent via QGroundControl. The base computer running this software can use either Linux or Windows due to QGroundControl's cross-compatibility. Most tests were conducted using an Arch Linux-based laptop; however, for this ROV, a Windows computer is recommended due to additional camera control software that can be utilized. The additional software for the gimbal camera is called Dragonfly, which can be found on the manufacturer's website. It allows for additional camera features to be used, including integrated AI object tracking and night vision. The manufacturer also provides a custom version of QGroundControl for Windows with Dragonfly integration; however, this was not tested due to operating system incompatibility [25].

C. Material Selection

The materials used in this design were selected based on factors like cost, accessibility and suitability for the project. For the main pressure hull the design used a pre-existing pressure hull product from BlueRobotics. This 3-inch diameter pressure hull had been pre-tested and came with assembly instructions. The datasheet specifies a maximum depth for each component, with 100 meters being the max depth for the

acrylic hull. This selection was crucial and custom designed pressure hulls were deemed to be too time-consuming and expensive for this project. The key frame material used was Item aluminium profiles, these are light-weight, waterproof and can be ordered in custom lengths. Another strength of using Item aluminium profile is the designed grooves which allows for seamless attachment of additional components to the frame. For external mounting to the hull, BlueRobotics clamps were secured around the cylinder to ensure the pressure hull remains rigidly mounted to the frame. To secure these clamps and other external connections to the aluminium frame, marine-grade 316 stainless steel bolts and screws were utilized. The selection of stainless steel was critical to minimize galvanic corrosion between the fasteners and the aluminium frame, thereby preventing rust and ensuring the structural integrity of the vehicle during underwater operations. Custom 3D-printed mounts were also developed to secure auxiliary components, such as the Z-1Mini micro gimbal camera. The materials used for 3D printing were PLA and water-based resin, which were chosen as the most cost-effective, entry-level materials for printing. Finally, EVA foam was selected as the buoyancy material; chosen for its high commercial accessibility and low density of 85 kg/m^3 , it performed exceptionally well and was easy to customize.

D. Electronic System Selection and Design

During the development of the ROV, the system architecture was modified multiple times, initially using an onboard companion computer like a pi4 as the main interface between the base station and the flight controller. However, this setup had to be modified due to size constraints. Below details the final components and architecture used in the ROV.

1) *Payload:* To demonstrate the capabilities of the ROV, a high-powered underwater light from BlueRobotics was selected as the mounted payload. The use of the Item aluminium frame on the bottom allows for future custom payload mounts to be designed without large frame design changes. This comes with the caveat that for larger payloads a recalibration of the buoyancy elements may be required.

2) *Power Distribution and Safety Control Loop:*

To handle the high-current demands of the thrusters while maintaining system safety, a dedicated power distribution and switching architecture was designed. The primary power source is a Tattu 2200mAh 4S LiPo battery, providing a nominal 14.8V. To prevent high-current arcing across the main external power switch, a low-voltage control loop is implemented. This is achieved as a small auxiliary BEC steps the battery voltage down to 12V to power a Blue Robotics external switch. When this switch is toggled on, it energizes the coil of an automotive relay, closing the high-power circuit. Once the relay is energized, raw battery power flows through a Holybro PM02 power module, which provides critical voltage and current telemetry to the flight controller via serial communication, while stepping down the voltage to 5V to power the Pixhawk. The switched 14.8V line then feeds into a Matek Systems Power Distribution Board (PDB). The PDB distributes raw battery power to the high-draw thrusters, lights, and camera, while utilizing its integrated regulators to supply 12V to the onboard Fathom-X board and 5V to the Auvideo M80 Ethernet module.

3) *Flight Control and Propulsion:* The core flight control is managed by a Holybro Pixhawk 6X Mini flight controller running the ArduSub open-source firmware, the main interface with this is through the Pixhawk's Ethernet port. The Pixhawk reads real-time IMU sensor data to run stabilization loops and execute pilot commands. The propulsion system consists of four APISQUEEN U2 Mini underwater thrusters arranged in an underactuated configuration (two horizontal, two vertical). The Pixhawk controls these thrusters using pulse-width modulation (PWM) signals sent to the thrusters' integrated electronic speed controllers (ESCs), allowing for dynamic control of depth, yaw, surge, and pitch/roll stabilization.

4) *Payload and Video Integration:* The primary payload consists of an XF-Z-1MINI gimbal camera inside the housing and a Blue Robotics Lumen Subsea Light mounted to the frame below the housing. The camera captures high-definition video and streams it directly to the surface via RTSP over the Ethernet switch. Additionally, the camera's three-axis stabilization gimbal is con-

trolled by the Pixhawk via serial (UART) commands. The Lumen light provides high-intensity illumination for underwater visibility, with its brightness managed directly by the Pixhawk via a PWM signal.

5) *Communication Architecture*: The communication architecture follows a tethered model. The primary data link is a tethered connection via a Fathom-X tether interface board system, which provides high-bandwidth, low-latency communication between the ROV and the surface control station. At the surface, a base computer running QGroundControl is connected via USB to a handheld game controller for manual operation, and via Ethernet to a Fathom-X surface board. This board translates standard Ethernet signals into a two-wire differential signal for transmission down the 50m slim Fathom ROV tether. Onboard the ROV, a matching Fathom-X ROV board receives the signal and converts it back to standard Ethernet. This Ethernet line is routed to an Auvideo M80 Ethernet module, which acts as an onboard network switch, distributing network traffic to the companion interfaces and the IP camera.

6) *System Architecture*: A system diagram found in 2 shows the complete system architecture of the ROV. Note that ground connections have been omitted from this diagram.

IV. IMPLEMENTATION & PROTOTYPING

A. Design Phase

The project started using the BlueROV2's design layout and internal components as a baseline. This was selected as the BlueROV2 is a well documented and proven platform, which made it an ideal candidate for a starting point for this project. The task requirements also specifically stated that the BlueROV2 was too large, making its profile the maximum size allowed for this project, because of this the design was modified with downsizing being the main objective. The first modification was removing the external battery that the BlueROV2 carries below the main tube and modifying the internals to include the battery. This was done to reduce the overall profile of the ROV. The next step was to remove the buoyancy foam around the BlueROV2 and design a bare thruster layout, with future plans to use simulations to optimize the additional buoyancy and the

placement of the thrusters. Figure 3 is the initial design layout of the modified BlueROV2 to the left and its comparison to the original BlueROV2 to the right found in Figure 4.

This initial design was then taken to the project's main stakeholder for evaluation with the result being a series of design improvements. The stakeholder felt the overall profile was too large and needed to be reduced by around half the size. This engineering challenge led to a second version being designed with modifications to the external frame. The feedback was identical and it was clear that the internal design needed to be overhauled completely to meet the small form factor criteria. It was decided that to meet the tight space constraints and off-the-shelf component limitations, it would be best to remove as much internal components from the design. This included:

- All internal ESCs removed, with plans to use external ESCs attached to the thrusters.
- The main companion computer, with plans to rely on inputs from the tether system.
- Internal wiring management components, with plans to design custom connections for the modular frame system.

This new design also introduced a key limitation to the design, with two thrusters being removed to save internal wiring space and external mounting space. This change pivots the design to an underactuated ROV system, relying on 4 thrusters instead of the standard 6 thrusters found in the BlueRov2 design. The design was now supported by an experimental thruster configuration seen in Figure 5 and a reduced bare-bones internal layout seen in Figure 6.

This design was taken to the project's main stakeholder for evaluation, the key modifications and their ramifications were explained, and the design was approved for fabrication.

B. Production Phase

1) *Development Roadblocks*: Several key challenges and design revisions emerged during the early prototyping and assembly phases, primarily relating to spatial constraints and electrical integration:

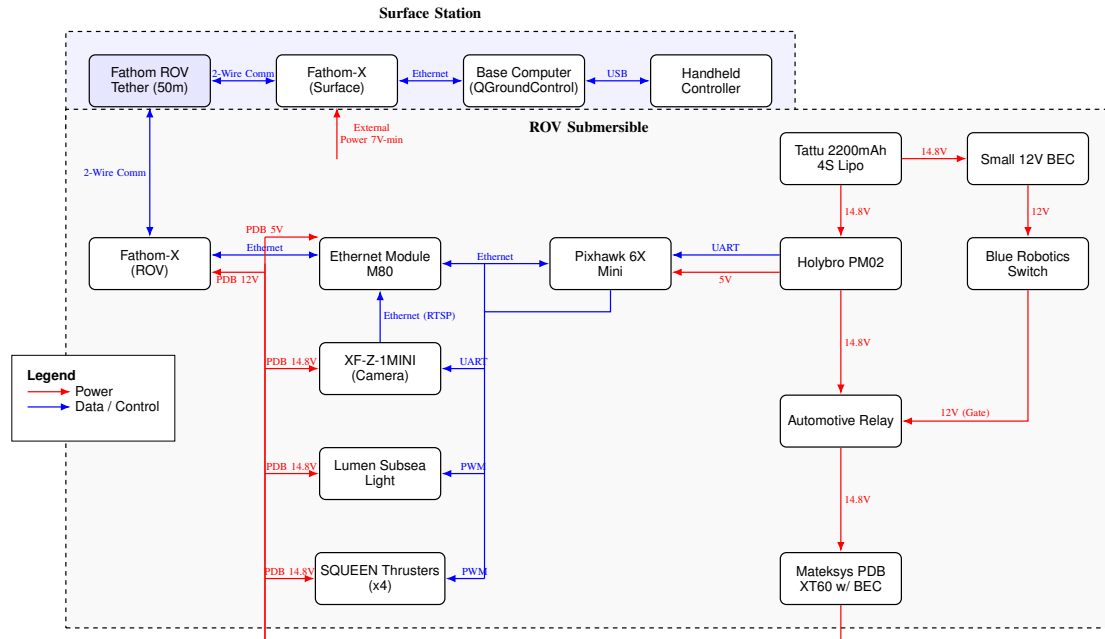


Fig. 2. System Architecture and Wiring Diagram for the SubbyROV, illustrating the power distribution and data bus routing between the Pixhawk, Fathom-X tether, and auxiliary payloads.

- 1) **Thruster Configuration Reversion:** The initial, experimental vectored thruster layout lacked empirical validation and was not supported by simulation or physical testing. Given the complex hydrodynamics, tight project timelines, and a lack of established literature for this specific geometry, the experimental configuration was abandoned. Instead, the design reverted to a conventional layout comprising two vertical and two horizontal thrusters. This configuration is well-documented in literature and natively supported by open-source control software like QGroundControl, thereby mitigating control system development risk.
- 2) **Spatial Constraints and Component Layout:** During early physical assembly, major spatial conflicts arose within the pressure hull. The initial CAD layout did not account for the volume occupied by internal wiring harnesses, component connectors, and bulky thruster cables. Consequently, the open-source BlueROV2 internal mounting plates were replaced by a custom, iteratively

designed 3D-printed chassis to optimize internal space. This custom chassis resolved several secondary design discrepancies:

- *Battery Downsizing:* The battery was sequentially downsized from a 29,000 mAh Lithium-Ion pack in the conceptual design, to a 5,200 mAh LiPo pack, and ultimately to a compact 2,200 mAh LiPo pack to fit within the hull.
- *Video Transmission:* The camera could not stream video directly through the Pixhawk flight controller as initially planned, requiring a dedicated Ethernet connection.
- *Relay Control Voltage:* The automotive relay used for the power switch logic had a control voltage limit 2 V lower than the battery's nominal voltage, necessitating an auxiliary voltage step-down BEC to prevent coil damage.

Resolving these spatial and electrical limitations was a critical prerequisite before buoyancy cal-

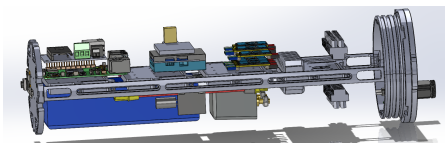


Fig. 3. Initial Design Layout of the modified BlueROV2, showing the removal of the external battery tube to reduce the vehicle's overall vertical profile.

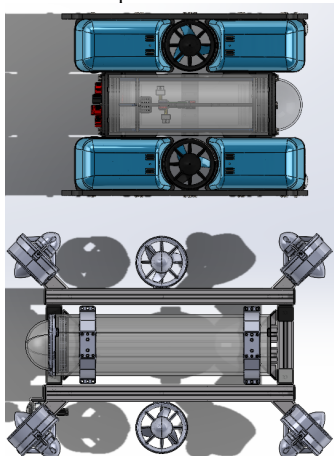


Fig. 4. Size comparison between the initial modified design (bottom) and the original BlueROV2 (top), highlighting the need for further miniaturization to meet the project's spatial constraints.

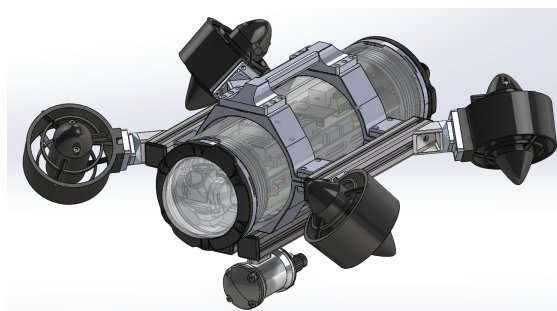


Fig. 5. Experimental 4-thruster vectored configuration proposed for the Version 2 design, aiming to provide coupled control with a minimal footprint.

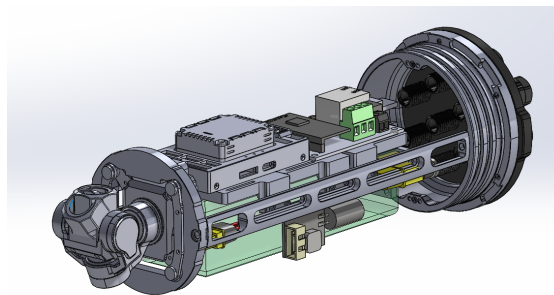


Fig. 6. Reduced barebones internal layout for the Version 2 design, eliminating internal ESCs and the companion computer to fit within the 3-inch pressure hull.

culations and hydrodynamic simulations could be performed, which in turn delayed the finalization of thruster positioning and buoyancy foam distribution.

To solve these issues the BlueRobotics internal mount was removed, and a custom internal mount was designed. Along with physical prototyping this custom mount was iterated several times and finalized when all components could fit inside. Figure 7 and Figure 8 show the updated internal mount and the physical construction of the new internal mount.

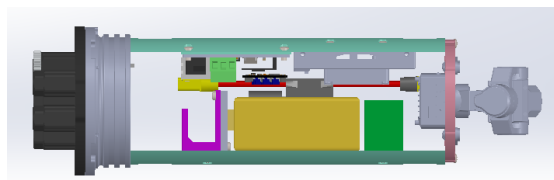


Fig. 7. Updated 3D-printed internal mount designed for Version 3, iteratively refined to safely house the downsized LiPo battery, Pixhawk, and power distribution electronics.

2) *Buoyancy and Stability Design:* Once all the internal components were fitted the model was ready for calculating the required buoyancy foam to offset the weight of the ROV to create slightly positive net buoyancy. First the weight and center of mass of all the internal components were calculated using SolidWorks Mass Properties, then with the external frame. The calculated weight allowed for the calculation of the required volume of buoyancy foam needed to offset the weight of

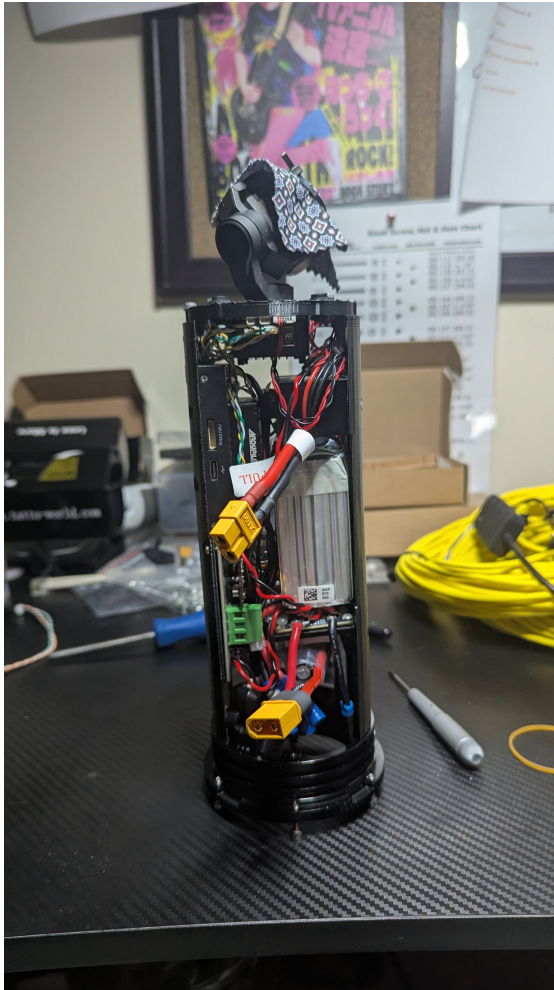


Fig. 8. Physical assembly of the Version 3 internal mounting chassis, fabricated from PLA to securely integrate the electrical components within the pressure hull.

the ROV. The additional weight from the foam and its mounting solution were then added to refine the calculation one more time. Once the volume of foam was calculated the handle part in the original design was modified to hold the required foam. Due to the complexity and size of this part it was printed with a resin printer, which was not a part of the original budget or design. The results of this change will be discussed further, however the part needed to be calibrated to ensure the COB was above the COM and aligned in all other axis. This was an important requirement as the passive

restoring force from this was required for ROV operation. SolidWorks was used to calculate the center of buoyancy by solving the COM of the volume of water displaced by the full ROV. However before this was fabricated the team wanted additional validation of the buoyancy design.

The design was exported into an STL, where a profile based on its mass properties was defined and a simple buoyancy environment test was created in the Stonefish ROS2 package. This test was run to validate the buoyancy design and to ensure that the ROV was stable in water; the simulator can be seen in Figure 9. The results showed that the COB was too far forward and the ROV would pitch up when floating. In Stonefish the mass parameters were then modified based on qualitative results from the simulator until a desirable transient response was achieved. These tweaks were then applied to the SolidWorks model, which then was sent back to the simulator for evaluation. This iterative process was repeated until the native mass properties produced an ideal response. Once this was confirmed the mount was fabricated and can be seen attached in Figure 10.



Fig. 9. Buoyancy test results simulated in Stonefish for the initial handle design, revealing an improper center of buoyancy that caused the ROV to pitch upward.



Fig. 10. Physical assembly of Version 3 featuring the custom resin-printed buoyancy handle, designed to correct the center of buoyancy and provide a passive restoring force.

C. Design Specification

After the buoyancy calculations were finalised the design was ready for testing. The final parameters can be found in Table I. While Table I highlights the costs of key components, a comprehensive full project budget and Bill of Materials (BOM) can be found in Appendix VIII-B.

The SolidWorks drawings of the full assembly, internal assembly and BOM are attached in Appendix B with the Figures 11 and 12 showing the SolidWorks Renders compared to the BlueROV2 [26], these figures visually show the size difference of the two ROVs.

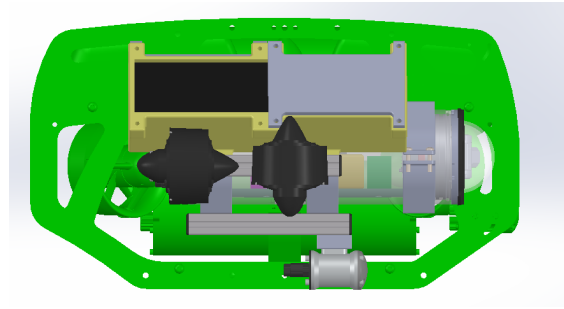


Fig. 11. Side-profile comparison rendering of the full SubbyROV assembly against the original BlueROV2 [26], demonstrating the significant reduction in overall footprint.

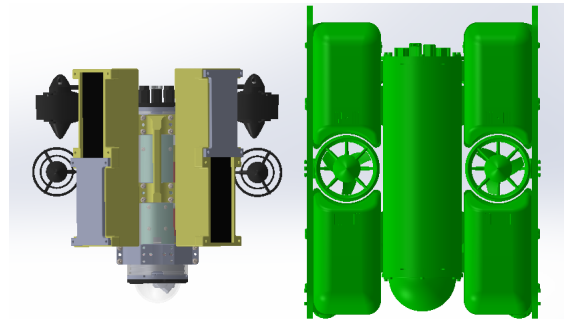


Fig. 12. Top-down comparison rendering of the SubbyROV and the BlueROV2 [26], emphasizing the reduced width and compact modular frame.

V. TESTING & RESULTS

1) Thruster, Payload and Kill Switch Testing: The first test was thruster control via the QGroundControl software. The experimental setup can be seen in Figure 13. The ROV was powered via an external power supply and the thrusters were connected to the power distribution board, the PWM signals were connected to the Pixhawk and the Pixhawk was connected to the base computer via USB. This allowed for thrusters to be powered and tested via QGroundControl and the configuration and PWM signals could be switched and modified to match the configuration. Care was taken to not fully power the thrusters on land as this could damage the motor. The high-powered flashlight payload was also tested in an identical manner and could be controlled via a switch on the

TABLE I
FINAL DESIGN AND MASS PARAMETERS OF THE SUBBYROV

Parameter	Value
Dimensions & Volume	
Length	308.90 mm
Width	362.26 mm
Height	232.88 mm
Displaced Volume	4.833 L
Mass & Buoyancy	
Calculated Dry Weight (SolidWorks)	4.568 kg
Actual Dry Weight (Physical)	4.750 kg
Calculated Buoyancy (Freshwater)	4.833 kg
Net Buoyancy (Calculated)	0.083 kg
Center of Mass (Origin @ COM on SolidWorks file)	
COM (X_G, Y_G, Z_G)	(0.00, 0.00, 0.00) mm
Moments of Inertia (at COM)	
I_{xx}	32,530,093.03 g · mm ² (0.03253 kg · m ²)
I_{yy}	35,146,249.06 g · mm ² (0.03515 kg · m ²)
I_{zz}	53,369,738.41 g · mm ² (0.05337 kg · m ²)
I_{xy}	950,747.74 g · mm ² (0.00095 kg · m ²)
I_{xz}	692,473.29 g · mm ² (0.00069 kg · m ²)
I_{yz}	-487,455.42 g · mm ² (-0.00049 kg · m ²)
Financials & Key Component Costs	
Fathom-X Tether Interface (Qty 2)	\$424.30 AUD
Watertight Enclosure (3" set)	\$417.57 AUD
XF-Z-1MINI PC Board	\$363.98 AUD
Fathom ROV Tether (50m)	\$303.43 AUD
Pixhawk 6X Kit	\$300.00 AUD
APISQUEEN U2 Mini Thrusters (Qty 4)	\$254.60 AUD
Lumen Subsea Light	\$212.42 AUD
Total Estimated Cost (BOM)	\$2,884.38 AUD

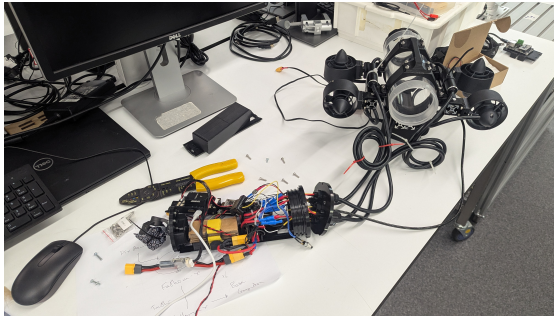


Fig. 13. Experimental setup for initial dry lab testing, enabling validation of the power distribution, PWM signal routing, and QGroundControl teleoperation before hull integration.

handheld controller. Once the thrusters were confirmed to be working, the custom kill switch was then added between the PM02D and the power distribution board. The switch was tested via a

multimeter to ensure that all voltage and current levels were at expected levels. This test revealed a flaw with the power flow design: when the kill switch is turned off, the design's goal was for power to be cut to all high-powered electronics. Because this interruption occurs after the PM02, the Pixhawk remains powered, allowing the battery level to be monitored. While this functioned flawlessly in testing, in the finalized design, the tether system (the ROV's communication link to the surface) is powered by the PDB, meaning this monitoring feature is lost upon triggering the kill switch. This is an acceptable trade-off for the safety and simplicity of the design.

2) *Simulation Testing:* Two types of simulation were done during the scope of this project. The first and main simulation was led by Brad and used the software package Stonefish. A secondary supporting simulation was performed in MATLAB to model the underactuated dynamics while the

Stonefish package was still being designed. This was conducted to validate the underactuated design choice.

a) Matlab Underactuated LQI design Study:

To validate the underactuated design choice, a comprehensive simulation study was conducted in MATLAB to test the stability and control of the ROV architecture. The study derived a 3-DOF planar model and implemented a discrete-time Linear Quadratic Integral (LQI) controller with an anti-windup scheme. The results demonstrated that this control architecture provides robust waypoint tracking, successfully compensating for non-linear quadratic drag, actuator saturation, and severely mismatched dynamics. It must be distinctly noted that this simulation utilized a generalized ROV model with differing physical parameters rather than the exact specifications of the ROV constructed in this project. This abstraction was necessary due to missing parameter data in the current design, specifically the complex hydrodynamic coefficients required to formulate the non-linear system of equations. Furthermore, the study established a critical operational requirement for this control methodology: the LQI controller relies on full state feedback. For this approach to be physically valid, the ROV must have accurate, real-time measurements of both absolute position and depth, as relying on dead reckoning leads to terminal position drift. Despite these limitations, the study serves as a robust proof-of-concept. The successful stabilization of the complex underactuated plant mathematically proves that this LQI control approach is highly resilient and can be directly applied to control the fully designed ROV in future operations once the necessary sensory and hydrodynamic data are acquired. This robust performance is visualized in Figure 14 and Figure 15. Information about this work can be found in the open-source SubbyROV repository [27]

b) Stonefish Dynamics Study: Building upon the initial buoyancy validation, the Stonefish simulation environment was extensively expanded by the simulation engineer to perform a comprehensive underwater dynamics study. The physical properties of the ROV, including its mass, moments of inertia, and center of gravity, were accurately modeled within the simulation utilizing

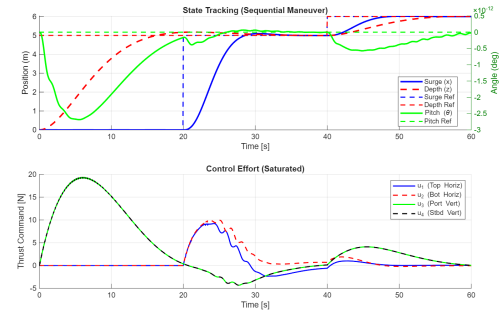


Fig. 14. Simulation results demonstrating robust LQI tracking performance on the true Non-Linear plant.

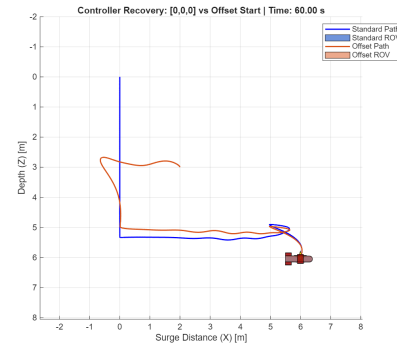


Fig. 15. Simulation output diagram showing the underactuated ROV moving between waypoints. This graph contains 2 simulations, with one starting at 0,0,0 and the other at a random initial position

the exported meshes. Custom ROS2 nodes were developed to interface with the simulated vehicle, enabling direct manual control via teleoperation scripts and the application of discrete "step inputs" to the thrusters. Through these inputs, the team evaluated the complex underactuated thruster dynamics and overall vehicle stability. A dedicated IMU data logging node was employed to monitor the real-time kinematic response, measuring angular velocities, linear accelerations, and calculating the net forces acting on the body frame, as seen in Figure 16. This high-fidelity testing allowed the team to observe how the four-thruster configuration behaves dynamically before physical deployment, providing a deep understanding of its

maneuvering characteristics and structural stability under varied control inputs. A visualization of this advanced simulation environment in action is presented in Figure 17.

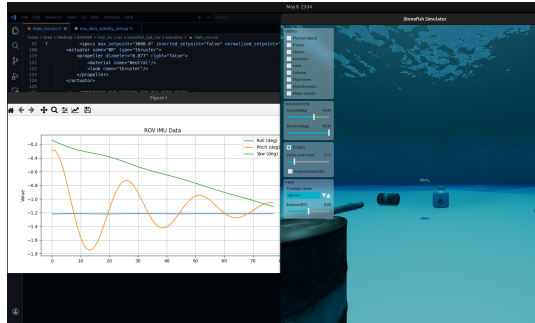


Fig. 16. Real-time kinematic response logging in the Stonefish simulator, extracting simulated IMU data to analyze angular velocities and linear accelerations.

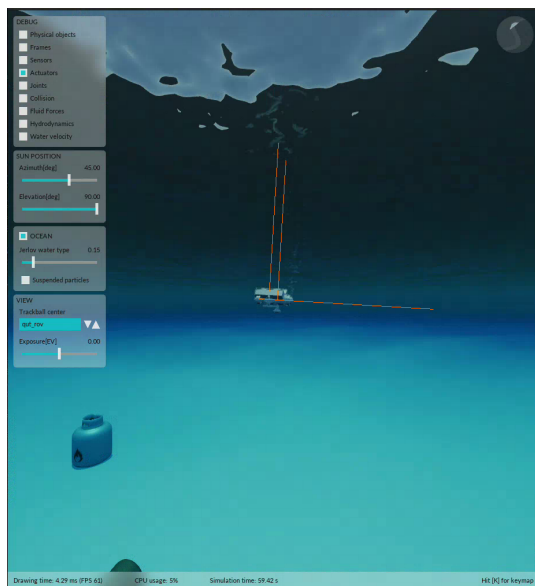


Fig. 17. The expanded Stonefish simulation environment used for testing thruster dynamics and vehicle stability under manual step inputs.

3) *Physical Testing:* Due to aforementioned roadblocks with the design, a fully measured physical test was no longer possible. To still meet project requirements, the physical test site was moved to a private pool. This test site had

similar hazards to the original test site as can be seen on the testing Risk Assessment found in Appendix VIII-E, however, these hazards could be controlled much more effectively than those at the original test site while still getting a test in the water. Once simulations were completed and the design was validated, the team moved onto physical testing of the ROV at the new site. Due to time constraints, only two simple tests occurred in the water: the first being a waterproof test, and the second being two manually controlled mobility tests in the water. The waterproof testing was conducted first, where the ROV's internals were removed and the chassis was sealed shut. The chassis was then submerged for a period of time to ensure that water did not ingress into the chassis. When the test was complete, the internal hull was checked for any water, and it was found to be completely dry. With this test completed, the manual control tests could take place. The goal of these tests was to qualitatively assess the dynamics of the ROV and to observe how the four-thruster configuration behaves dynamically, providing an understanding of its maneuvering characteristics and structural stability. These results could then be compared to the Stonefish simulation, and if the digital twin accurately represented the physical ROV, this would validate the simulation and allow it to be used for more advanced control development. Due to time constraints and lack of measurement resources available at the test site, an unmeasured starting point was used, and no set path could be implemented. This meant that measured IMU pitch data found in Figure 18 cannot be compared directly to the simulation data; however, it does show that in future tests the ROV has the capabilities to produce comparable quantitative results.

Figures 19, 20 and 21 show the ROV submerged underwater under manual control. Using a specialised phone case, underwater videos were captured during these tests. The first test occurred after dark and showcased the powerful flashlight payload in action along with very basic operational capabilities. The second test focused on demonstrating the current mobility of the ROV and how it compares to the simulation. Operation occurred through line of sight with the camera stream being presented on the control computer



Fig. 18. IMU pitch log recorded during Test 1, capturing the physical vehicle's pitch response during unmeasured manual control inputs in the pool environment.

as additional guidance.

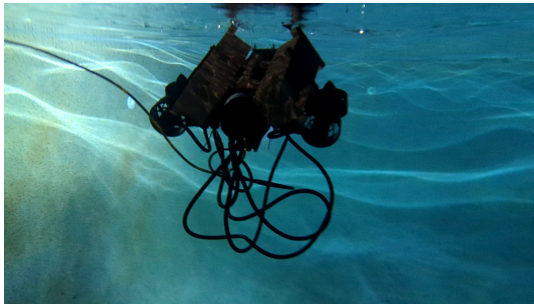


Fig. 19. SubbyROV operating submerged during Manual Test 1. The environment is actively illuminated by the high-intensity Lumen Subsea Light payload.

While the first manual test was shorter and was conducted in the dark to show the basic capabilities of the ROV, the second ran for roughly 10 minutes and consumed the charge of the LiPo battery down to 10 percent. This revealed the ROV's standard operational time and that extended use of the high power payload significantly affects operational time. Test videos can be located in the SubbyROV repository [27].

4) *Testing Comparison:* When comparing the real word dynamic response to thruster step inputs (manual control) and the Stonefish simulations, distinct properties worth noting include:

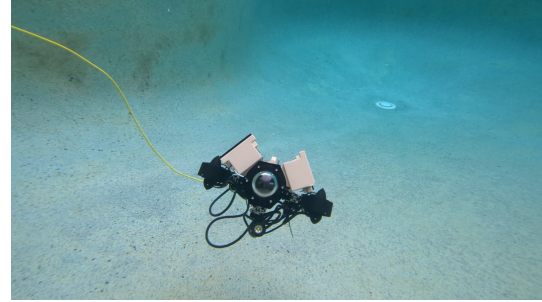


Fig. 20. SubbyROV during Manual Test 2, demonstrating basic maneuverability and structural stability under line-of-sight teleoperation.

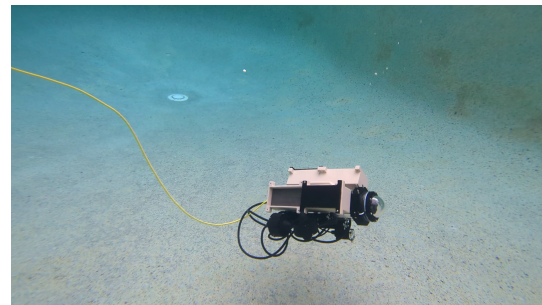


Fig. 21. Alternative view of the SubbyROV during Manual Test 2, confirming waterproof integrity and buoyancy balance in a controlled pool environment.

- **Stronger Buoyancy:** During operation, the ROV exhibited significantly greater buoyancy than anticipated. This was evident in its tendency to ascend faster when compared to simulations, 1 half of the layer foam inserts were removed to compensate for this.
- **COB Location Differences:** Viewing the response of the Real ROV it was clear the real COB was further forward when compared to the simulation. This was evidenced by the ROV sitting slightly nose up in the water.
- **Similar Dynamics:** While there were some slight differences between the simulation and real ROV properties, the test revealed that the simulator's prediction of how the ROV dynamically moves through the water was qualitatively similar to the real test, proving the simulations effective modeling of the underwater underactuated system.

VI. DISCUSSION AND PROJECT EVALUATION

1) *Safety and Risks:* Throughout the development and testing of the SubbyROV, strict adherence to safety and risk management was prioritized. A comprehensive risk assessment was developed prior to any physical construction or testing, outlining potential hazards such as electrical shock, manual handling injuries, and water-related risks. To mitigate environmental and public hazards during testing, operations were relocated from public open-water settings to a private pool. This environment allowed for strict control over the testing site while continuing to satisfy the safety guidelines. Electrical safety, specifically concerning the high-capacity Lithium-Polymer (LiPo) batteries, was a critical focus. Following safety protocols, a specialized LiPo-safe battery bag was utilized at all times during transport, storage, and charging. The only instance the battery was removed from this fire-retardant bag was during active installation and operation within the ROV's sealed pressure hull. Looking forward, the overall risk profile for future work and continuous use of this ROV remains manageable but requires diligence. Future developments such as the integration of advanced sensors or autonomous algorithms will introduce new complexities, particularly regarding the handling of ultra-low voltage electronics and the prevention of water ingress into newly added components. Operators must ensure ongoing maintenance of the waterproof seals, proper storage of batteries, and adherence to standard operating procedures to mitigate risks of electrical failure or loss of the vehicle. The complete approved Risk Assessment has been included in Appendix VIII-E for reference.

2) *Environmental Impact:* Environmental considerations and the sustainability of the materials used were important throughout the development process. The majority of the frame and components for the ROV were 3D printed using PLA (Polylactic Acid) filament for prototyping. PLA is a bioplastic derived from renewable resources such as corn starch or sugarcane, making it a more environmentally friendly alternative to traditional petroleum-based plastics [28]. Its biodegradability under specific industrial composting conditions further reduces its long-term environmental im-

pact. However, the current biodegradability is limited in natural environments, persisting in landfills for decades if not properly composted. The choice of PLA was primarily driven by its ease of use in rapid prototyping and its suitability for printing the custom structural components required for the ROV. While not as durable or water-resistant as materials like ABS or PETG, PLA provided a balance of accessibility, cost-effectiveness, and sustainability for the scope of this project. Electrical components and wiring were sourced from existing electronics and surplus materials where possible, following a reduce and reuse philosophy to minimize waste. The use of off-the-shelf components aligns with a sustainable design approach by leveraging existing technologies rather than developing new, resource-intensive solutions from scratch. Moving forward, opportunities exist to further enhance the environmental sustainability of the SubbyROV. Exploring alternative biodegradable filaments with improved mechanical properties or water resistance could provide a more durable yet eco-friendly option. Additionally, incorporating sensors and components made from recycled materials or implementing a comprehensive end-of-life recycling program for the ROV would contribute to a circular economy model. Overall, the project demonstrates that cost-effective, small-scale robotics can be developed with a conscious consideration for environmental impact.

3) *Limitations and Weaknesses:* There are some limitations that need to be acknowledged when looking at the results of this project. These limitations were driven by a number of factors including time constraints, the size of the team working on the project and the hardware limitations of the off-the-shelf components used. Listed below are key limitations that must be considered before the result is evaluated:

- **External Sensing:** The only sensors on the design are the internal sensors located in the camera gimbal and the Pixhawk controller. There is currently no external depth sensor, Doppler velocity log sensor, or sensor that can capture position data. This means further modifications must be made for fully autonomous operations using standard techniques.

- **Operational Time:** The current operational time is roughly 10 minutes based on a single test. While this was deemed acceptable for the project due to easy battery accessibility for manual recharging, it is not suitable for long term or repeated operations and would need to be significantly improved.
- **Thruster Configuration:** The thruster configuration will require additional control effort and this still won't remove all effects of underactuation perfectly. The result of this was seen in the MATLAB simulation as surge movements cause the pitch angle to oscillate, which would ruin any undampened camera recordings.
- **Material Limitations:** The 3D printed PLA components are not suitable for long-term submersion in water and will need to be replaced with a more appropriate material such as fabricated aluminium parts. This was apparent as the water-based resin buoyancy handle warped around the edges after drying, and PLA mounts would split over continuous use without proper printing properties.
- **Operational Limitations:** As the only source of communication with the Pixhawk is the tether, any additional inputs for using an EKF are traditionally done with a Pi communicating over UART need to be modified. One example of this would be processing the camera stream received on the base computer, then sending commands back via the tether. This process would need to be custom built and tested as current software would not have this feature built in.
- **ArduSub Implementation:** The ArduSub software stack is well suited for this project, however, due to the time constraints of this project, only basic customisation was conducted. It is expected that future modifications to the ROV design would require further software customisation which may not be well documented due to experimental use of the software as the main controller.
- **COTS Components:** The reliance on COTS components, while beneficial for rapid development, introduces dependencies on manufacturer specifications and support. The effect of this was evident as the pressure housings

maximum ingress holes was 7, this limit the amount of external components that could be added to the ROV design, including 2 additional thruster mounts or external sensors such as a depth sensor or doppler velocity log. A custom plate with more holes may be required when increasing the DOF or adding additional sensors. This same issue occurs with the selected PDB, this board has 2 BEC slots(5V, 12V) and 6 14.8V slots which in the current design are all in use.

- **Pressure Housing:** The pressure housing sold by BlueRobotics recommended the use of a release valve which automatically regulates the difference between internal and external pressure. Due to ingress hole limits(7) and no specific depth requirements in the task criteria, the use of a release valve was omitted to save space. If future iterations of the project were to dive deeper, a release valve would be required to ensure the ROV can safely withstand the pressure changes.
- **Internal Temperature:** The effect of prolonged use and the internal components heating up was not measured or documented, if the components are used for prolonged periods it is unknown how this would effect the ROV. Internal sensors on the Pixhawk and the Z1-MINI can measure their board temperatures, so future performance tests on this may be required.

When reviewing the difference between the simulation test and the pool test, there were some noticeable differences. This parameter differences can be traced back to the below assumptions and idealisations

- **Missing Models:** When designing the 3D files for the final design, some elements were not modeled as no files existed or were not easily accessible. These include the ESC tubes attached to the Thrusters and all internal and external wires. The lack of these components in the CAD model and simulation results is evident in the pool test results in the way of a different COM and COB. This can be address via onsite modification of the ROV or extensive measurement and addition of missing models in the digital twin.
- **Weight Assumptions:** Using Solidwork's

built in density calculator is only an approximation of weight as it assumes materials are homogenous and uniform. The actual weight of each component may vary depending on the manufacturing process (e.g. 3D printing) and any additional parts added to the final design. A scale could be used to weigh each component more accurately

- **Idealistic environment parameters:** Parameters such as water temperature and water salinity were not factored into the simulation. In the pool test, the waters parameters were not considered or measured, in turn affecting the dynamics of the ROV.
- **3D Print Weights:** 3D print weights were calculated using the slicing software for each model and then added together. However, the weight of each part may vary depending on the manufacturing process (e.g. 3D printing) and any additional parts added to the final design.
- **Real and 3D Mismatches:** While the real ROV was designed to be 1-1 scale with the 3D CAD model, there were some unavoidable differences. These include slight angles changes and component mountings.

4) *Goals and Objectives:* After considering the results above, the goals and objectives of the project can be evaluated against the success criteria outlined in the project plan:

- **Goal 1: Cost Effective Small Scale ROV:** The project successfully delivered a cost-effective small scale ROV with off-the-shelf components. The solution could demonstrate manual operation, customisation, easy handling and payload integration while being simple enough to set up and operate.
- **Goal 2: Controlled Maneuverability:** The project delivered a solution that could demonstrate controlled diving and movement using a thruster configuration. With the caveat that these operations need to be supported by future control implementations for stable operation.
- **Goal 3: Autonomous Capabilities:** This has not been achieved yet due to the constraints of the project. Limiting factors which prevented this goal include lack of external sensing, limited processing capabilities for on-

board AI operations, and insufficient thruster authority for unsupported autonomous maneuvers.

- **Goal 4: Demonstrator Capabilities:** This goal was achieved; the project was able to produce a working ROV prototype that was capable of quick and easy setup. There are many possible academic uses of this solution ranging from simple demonstration to advanced research on underactuated control design, small scale AUV development, and low cost sensor development platforms.
- **Goal 5: Document Findings:** This goal was also achieved as all findings and documentation has been made available for open-source use. These can be found on the associated GitHub repository [27].

The result of testing revealed the ROV can perform basic manual operation and camera stream, which when compared to the custom Stonefish simulator designed for further control and autonomous work demonstrates the ROV behaves in a manner consistent with the simulation. The main difference between Overall, the project was successful in that it met most of the outlined goals and objectives. It also highlighted the importance of time and budget constraints on project scope, particularly when working in small teams or solo. The request for additional team support halfway through the project allowed for the completion of the electrical integration and documentation which would have otherwise not been completed, demonstrating the importance of seeking support when needed.

VII. CONCLUSION

The SubbyROV project successfully delivered a cost-effective, small-scale Remotely Operated Vehicle designed specifically for research and academic demonstration. By leveraging off-the-shelf components, custom 3D-printed PLA structures, and an iterative design process informed by robust simulations, the team overcame significant mechanical and electrical integration challenges. The transition to an underactuated four-thruster configuration not only reduced the vehicle's footprint and complexity but also established a unique platform for advanced control system research. Physical testing in a controlled pool environment

validated the waterproof integrity of the custom chassis and demonstrated the vehicle's maneuverability under manual teleoperation. While the project achieved its primary goals of creating a functional, modular testbed, it also highlighted key areas for improvement, particularly regarding operational time, material durability, and the lack of external state-estimation sensors.

Looking forward, the next major phase of the project will pivot from hardware development to advanced software and control implementation. This future work, which will be spearheaded by Brad, focuses heavily on establishing robust autonomous capabilities for the SubbyROV. A primary objective will be the integration of external sensors such as Doppler Velocity Logs (DVL) and high-resolution depth sensors to facilitate accurate state estimation and localization. Utilizing the digital twin developed in Stonefish and the theoretical groundwork from the MATLAB LQI study, Brad will continue the development and deployment of advanced control laws capable of handling the highly non-linear, underactuated dynamics of the ROV. Furthermore, future work will explore the implementation of Machine Learning (ML) pipelines and computer vision algorithms, leveraging the onboard IP camera to enable autonomous navigation, object detection, and path following. By building upon the solid mechanical and electrical foundation established in this project, these subsequent autonomous control developments will transition the SubbyROV from a manually operated prototype into a fully capable, intelligent Autonomous Underwater Vehicle (AUV).

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VIII. APPENDICES

A. Appendix A: Original Literature Review (from Project Proposal)

1) *Introduction:* Unmanned Underwater Vehicles (UUVs) have become indispensable tools for a vast range of applications, including oceanographic surveys, infrastructure maintenance, and military defence [29], [30]. As technology has advanced, these vehicles have evolved from simple teleoperated platforms into complex autonomous systems capable of executing sophisticated missions [29]. However, the literature reveals that a significant portion of research and commercial development has focused on large-scale, work-class vehicles or highly specialised Autonomous Underwater Vehicles (AUVs), which are often financially and logistically inaccessible for educational

institutions or rapid prototyping research [1]. This is due to the cost of setting up demonstrations with large scale ROVs and larger ROV designs call for larger scale prototyping solutions which may not be realistic. This has created a gap for small-scale, low-cost, and modular ROV platforms that are not only suitable for demonstration but are also powerful enough to serve as testbeds for advanced control and perception algorithms. The aim of this review is to explore the current state-of-the-art in small-scale ROV design by analysing key themes in existing literature, thereby identifying best practices and informing the design methodology for a novel research platform.

2) *Methodology and Scope:* The literature review was conducted using a systematic approach, focusing on peer-reviewed journal articles, conference papers, and technical reports from the last decade. The search was primarily conducted using the database IEEE Xplore, with keywords including “small-scale ROV”, “underwater vehicle design”, “ROV control systems”, and “modular underwater robotics”. The review is structured to cover key design themes, hardware and software selections, control system architectures, and system modelling approaches, providing a comprehensive overview of the current state of small-scale ROV research. The scope of this review is limited to reviewing current literature on ROV design methodology and hardware selections.

3) Key Design Themes:

a) *Chassis and Mechanical Design:* The physical architecture of an ROV is a foundational design choice that dictates its hydrodynamic performance, modularity, and durability. The literature presents a clear dichotomy between open-frame and enclosed, torpedo-style chassis designs. Open-frame designs, such as those seen in the KCROV [31], MarmarOV [32], and eROV [2], are praised for their modularity, allowing for easy mounting of sensors, payloads, and thrusters. This configuration simplifies maintenance and is highly adaptable for research platforms. In contrast, streamlined, enclosed hulls are often selected to minimise hydrodynamic drag for greater efficiency and speed, as demonstrated by the MASUV-1 [3] and X4-ROV [4]. The thruster configuration is critical for maneuverability, with designs ranging from underactuated 3-thruster systems providing

basic 3-DOF motion [5], to fully actuated 6 or 8 thruster systems enabling full 6 DOF control [6], [7]. The placement of thrusters in a vectored arrangement is common for achieving high agility, allowing for coupled control of surge, sway, and yaw from a set of horizontal thrusters [1], [8].

Open-frame designs, such as those seen in The literature presents a clear dichotomy between open-frame and enclosed, torpedo-style chassis designs.



Fig. 22. (a) Open-frame chassis (eROV) [2]



Fig. 23. (b) Enclosed hull design (MASUV-1) [3]

Fig. 24. Examples of ROV chassis designs: (a) open-frame and (b) enclosed hull

Open-frame designs, such as those seen in the MarmarOV and eROV , are praised for their modularity, allowing for easy mounting of sensors, payloads, and thrusters. This configuration simplifies maintenance and is highly adaptable for research platforms. Figure 25 illustrates the impact

Scheme	Num	Roll	Pitch	Advantages	Disadvantages
	5	✗	✗	Minimum number of thrusters	Underactuated Low maneuver capability Balancing problems Only one motor for the vertical thrust Central motor reduces payloads space
	6	✗	✓	Limited number of thrusters Good components disposition	Underactuated Critical balancing of roll DOF especially in the open configuration
	6	✓	✗	Limited number of thrusters Good components disposition	Underactuated Critical balancing of pitch DOF
	7	✓	✓	Minimum number of thrusters to control 6 DOFs	Asymmetrical configuration (not suitable for torpedo shape)
	8	✓	✓	Completely symmetrical 6 DOFs completely controlled	More thrusters than necessary

Fig. 25. Comparison of different thruster orientations and their impact on maneuverability. [33]

of different thruster orientations on maneuverability, highlighting the trade-offs between simplicity

and control complexity. Having a vectored thruster arrangement allows for more maneuverability with a smaller form factor. It is seen in figure 25, that the orientation of the upward facing thrusters can significantly impact the vehicle's ability to roll and pitch efficiently. This makes the amount and orientation of thrusters a key consideration as the goal for the QUT demonstration ROV is to have a small form factor, while still being able to perform complex maneuvers.

b) Material Selection and Waterproofing.: Material selection directly impacts the vehicle's weight, durability, and operational depth. For low-cost educational platforms like the ArduinoSub and the vehicle proposed by the authors of [34], PVC and acrylic are common choices due to their affordability and ease of fabrication. For more robust applications, designs progress towards aluminium frames and hulls [30], [31], with high-performance vehicles intended for deep-sea operation employing materials like carbon fibre and syntactic foam to achieve high strength-to-weight ratios [35]. Waterproofing remains a critical and persistent challenge. Standard solutions include O-ring seals for static enclosures and potted cable penetrators [6]. However, the literature presents several innovative approaches to overcome the limitations of these methods. The Jeff AUV uses a novel non-penetrative magnetic coupling to transmit torque from internal motors to external propellers, completely eliminating the need for a dynamic shaft seal [36]. The design by the authors of [37] also highlights custom-developed connectors as a significant improvement over permanently potting electronics, allowing for easier maintenance and repair. For the QUT demonstration ROV, waterproofing will be a key challenge due to its small form factor.

4) Hardware Selections:

a) Propulsion and Sensing.: The choice of thrusters and sensors is dictated by the vehicle's mission profile and budget. Propulsion systems range from modified low-cost (~\$50) bilge pumps in budget designs [38] to commercially available, high-performance brushless DC thrusters like the BlueRobotics T200 (~\$350 per unit), which are noted for their power and reliability [2], [8]. Sensor suites vary significantly; educational kits may only include a basic IMU and a camera [1],

while advanced research platforms like EVA and Mesobot are equipped with a comprehensive array of high end high-cost sensors, including Doppler Velocity Logs (DVLs) for measuring velocity relative to the seafloor (~\$12,500), USBL positioning systems for underwater localization (~\$3000), and high-resolution waterproofed cameras for visual navigation (~\$300) [11], [12].

b) Control Hardware.: The selection of on-board electronics reflects a trade-off between cost, computational power, and development complexity. At the low-cost end of the spectrum, microcontroller based systems are prevalent. The D20-ROV and ArduinoSub, for instance, use an Arduino Mega 2560 as the central brain, which is sufficient for basic manual control and sensor interfacing [5], [9]. For more advanced capabilities, single-board computers like the Raspberry Pi are frequently adopted, as seen in the BlueRov2 and PolROV designs, providing a Linux-based environment capable of handling networking, video streaming, and more complex algorithms [8], [10]. High-performance vehicles designed for significant autonomy, such as EVA and Mesobot, employ multi-computer architectures with powerful CPUs and dedicated GPUs (e.g., Nvidia Jetson TX2) to process large volumes of sensor data in real-time [11], [12].

5) Software and Communication Frameworks: The software stack and communication architecture are integral to the ROV's functionality. For complex, modular systems, ROS is the framework of choice, as seen in the Autonomous ROV for Marine Growth and the EVA hybrid vehicle [7], [11]. ROS is valued for its extensive libraries, hardware drivers, and robust inter-process communication, which significantly accelerates development time. On simpler platforms, custom firmware is often developed in C++ within the Arduino IDE or using Python on a Raspberry Pi [1]. Open-source firmware like ArduSub, used in the VITA1 prototype, provides a powerful and well-supported alternative to developing a control system from scratch [13]. Communication is almost universally handled via a physical tether for ROVs, which provides both power and a high-bandwidth data link. This link is typically Ethernet over twisted pairs, though high-end systems may use fibre optics for greater bandwidth and distance [35].

6) *Control System Architectures:* Control strategies for underwater vehicles have matured from direct teleoperation to sophisticated autonomous behaviours [5]. The foundational control method discussed in much of the literature is the PID controller, used for fundamental tasks like depth-hold and heading-hold in vehicles such as the PolROV and Ariana-I [6], [10]. These controllers are well-understood and relatively simple to implement. However, due to the highly non-linear dynamics of underwater vehicles, more advanced techniques are often required for robust performance, especially in the presence of external disturbances like currents [5]. To address this, some designs implement Sliding Mode Control (SMC) for its robustness to parameter uncertainties [5], or Fuzzy Logic controllers which do not require a precise mathematical model of the system [14]. The Kambara AUV represents the cutting edge, employing model-free reinforcement learning to allow the controller to learn the vehicle's dynamics through trial and error, avoiding the need for an explicit model altogether [39].

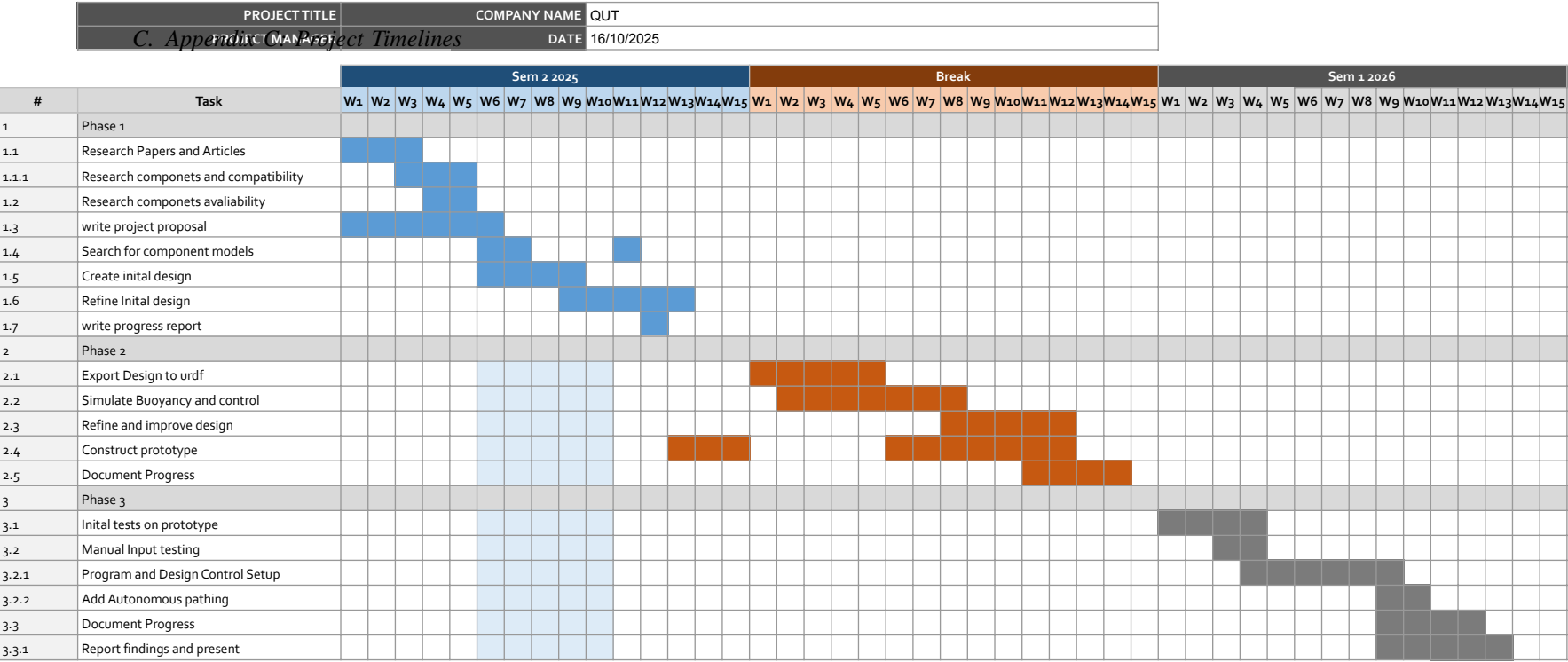
7) *System Modelling Approaches:* The development of a control system is often preceded by system modelling to simulate and predict the vehicle's behavior. A key distinction in the literature is the approach taken to derive this model. Many research-grade projects, such as the eROV and the spherical AUV by the authors of [40], develop a comprehensive 6-DOF dynamic model from first principles. This process involves using CAD software to determine rigid-body parameters (mass, inertia) and CFD simulations to estimate hydrodynamic coefficients (drag, added mass) [2], [8]. This model-based design allows for extensive simulation and controller tuning before physical construction. An alternative, empirical approach is system identification, where experimental data from a physical prototype is used to derive a transfer function model. This was successfully demonstrated by the authors of [41] for the UTERG-ROV2, providing an accurate model for depth control without complex theoretical analysis. The choice of modelling approach represents a trade-off between theoretical rigour and the practicalities of physical testing.

B. Appendix B: Comprehensive Bill of Materials (BOM) & Full Budget

TABLE II
SUBBYROV COMPREHENSIVE BILL OF MATERIALS (BOM) AND FULL BUDGET

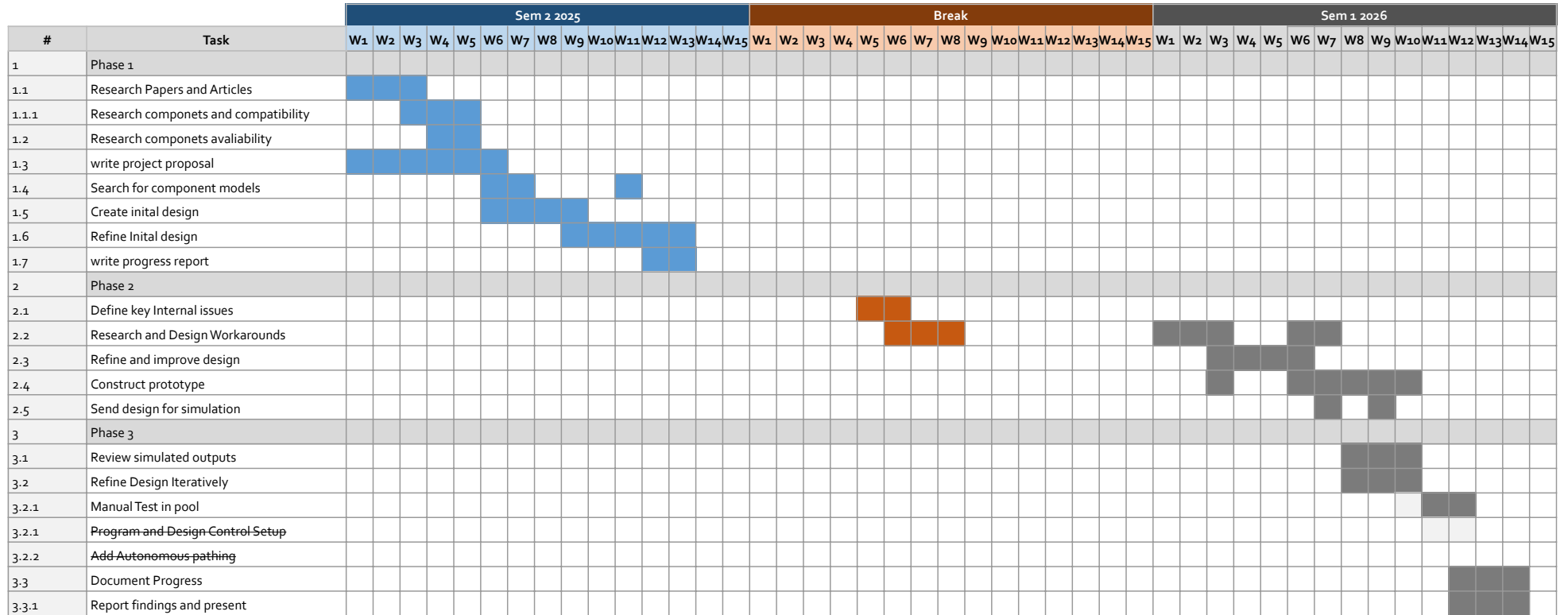
Sub-System / Part Name	Manufacturer	Part Number	Qty	Unit Cost (\$AUD)	Total Cost (\$AUD)	Onboard Wt est (g)
1. Propulsion						
APISQUEEN U2 MINI 1.3Kg Thruster	underwaterthruster	U2 mini	4	63.65	254.60	840
2. Electronics						
Pixhawk 6X + Mini Baseboard + PM02	Dr.Uav / HolyBro	6X-ICM-45686-5	1	300.00	300.00	50
Matek Systems PDB XT60 W/ BEC	dronesxpress	PDB-XT60	1	36.00	36.00	7.5
Fathom-X Tether Interface Board	Blue Robotics	BR-100178	2	212.15	424.30	100
Tattu 2200mAh 4s 45c Lipo Battery	NextFPV	TA52004S35	2	42.95	85.90	190
Durakool Automotive Relay SPDT 90A	Rs	915-6644	1	7.99	7.99	14
Switch	Blue robotics	BR-100433	1	42.43	33.94	20.5
XF-Z-1MINI	UmmanedRC	-	1	363.98	363.98	69
Ethernet Module M80	Auvideo	38658-RT	1	46.69	46.69	4.89
Assorted connections + Wires	Misc	-	0	0.00	30.00	0
Small 12V BEC	Cheapest	-	1	20.00	20.00	0
4. Frame & Enclosure						
Fathom ROV Tether	Blue Robotics	BR-100985-050	1	379.29	303.43	0
Watertight Enclosure 3 inch 240 set	Blue Robotics	WTE-VP	1	521.96	417.57	0
Flange	Blue Robotics	-	2	0.00	0.00	87
Enclosure	Blue Robotics	-	1	0.00	0.00	442
Dome	Blue Robotics	-	1	0.00	0.00	39
Plate	Blue Robotics	-	1	0.00	0.00	93
150mm Item Profile	Item	-	4	21.60	22.96	288
Item end caps	Item	-	10	0.95	9.50	0
T nuts Item	Item	-	20	3.65	73.00	40
Penetrators 7.5 HC(7.0+0.3mm)	Blue Robotics	BR-100870-175	4	19.74	63.17	87
Clamps	Blue Robotics	-	2	66.83	106.93	644
5. Buoyancy						
15mm EVA Foam Sheet, 1 x 1 Metre	Titan AV	-	1	22.00	22.00	122.4
6. Fasteners						
M3	-	-	10	0.00	0.00	0
M4	-	-	20	0.00	0.00	0
7. Payload						
Lumen Subsea Light	Blue Robotics	BR-100857	1	265.52	212.42	118
8. 3D Printed						
PLA filament	Anywhere	-	1	50.00	50.00	0
Internal Mounts A and B	Custom	-	2	0.00	0.00	72
Switch Mount	Custom	-	1	0.00	0.00	13
PDB Mount	Custom	-	1	0.00	0.00	13
Camera Mount	Custom	-	1	0.00	0.00	21
Side Item Mount	Custom	-	4	0.00	0.00	40
Thruster Mount A	Custom	-	2	0.00	0.00	22
Thruster Mount B	Custom	-	2	0.00	0.00	20
Handle	-	-	1	0.00	0.00	400
Light Mount	-	-	1	0.00	0.00	13
				TOTALS:	\$2,884.38	3,209.29 g

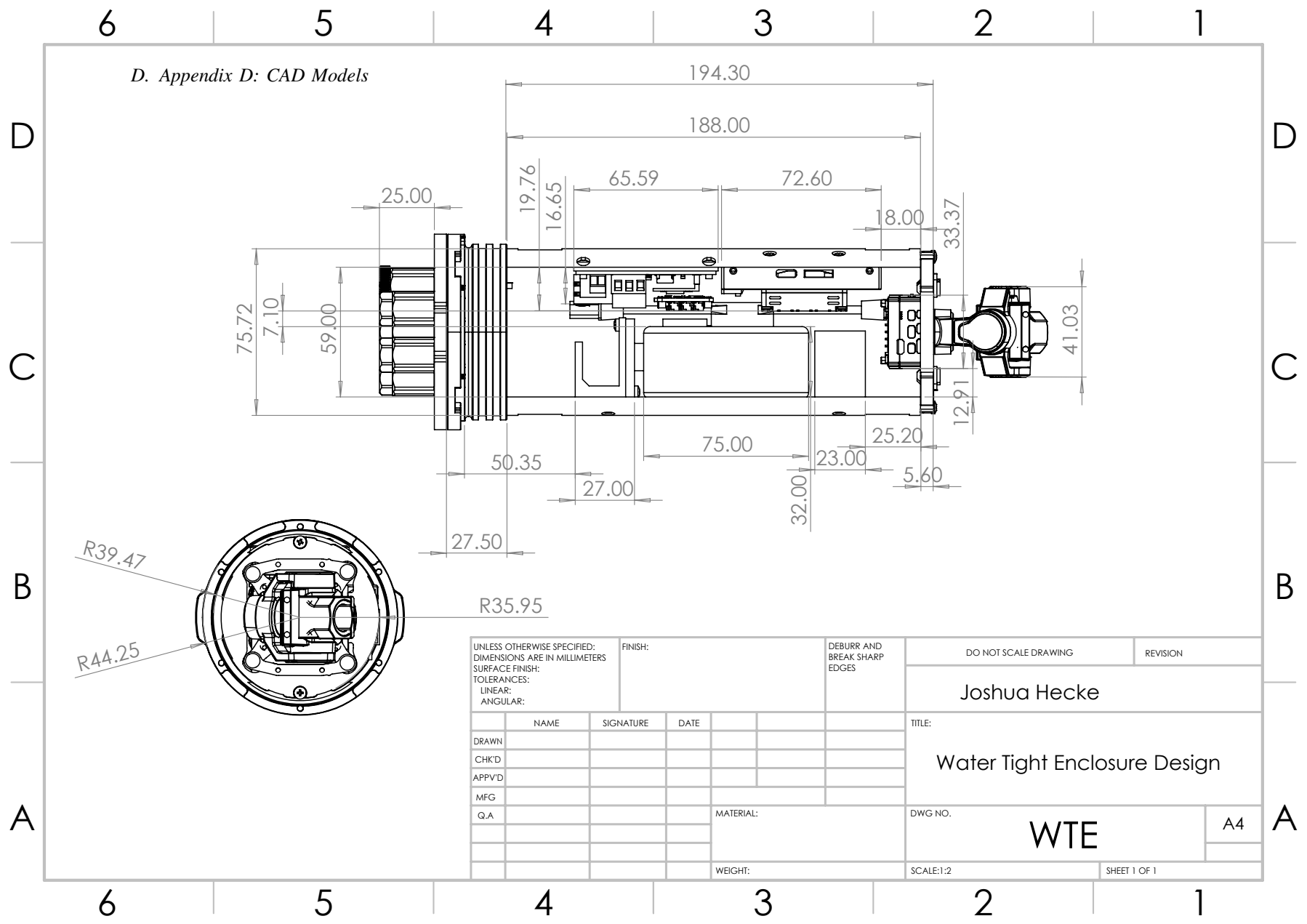
GANTT CHART

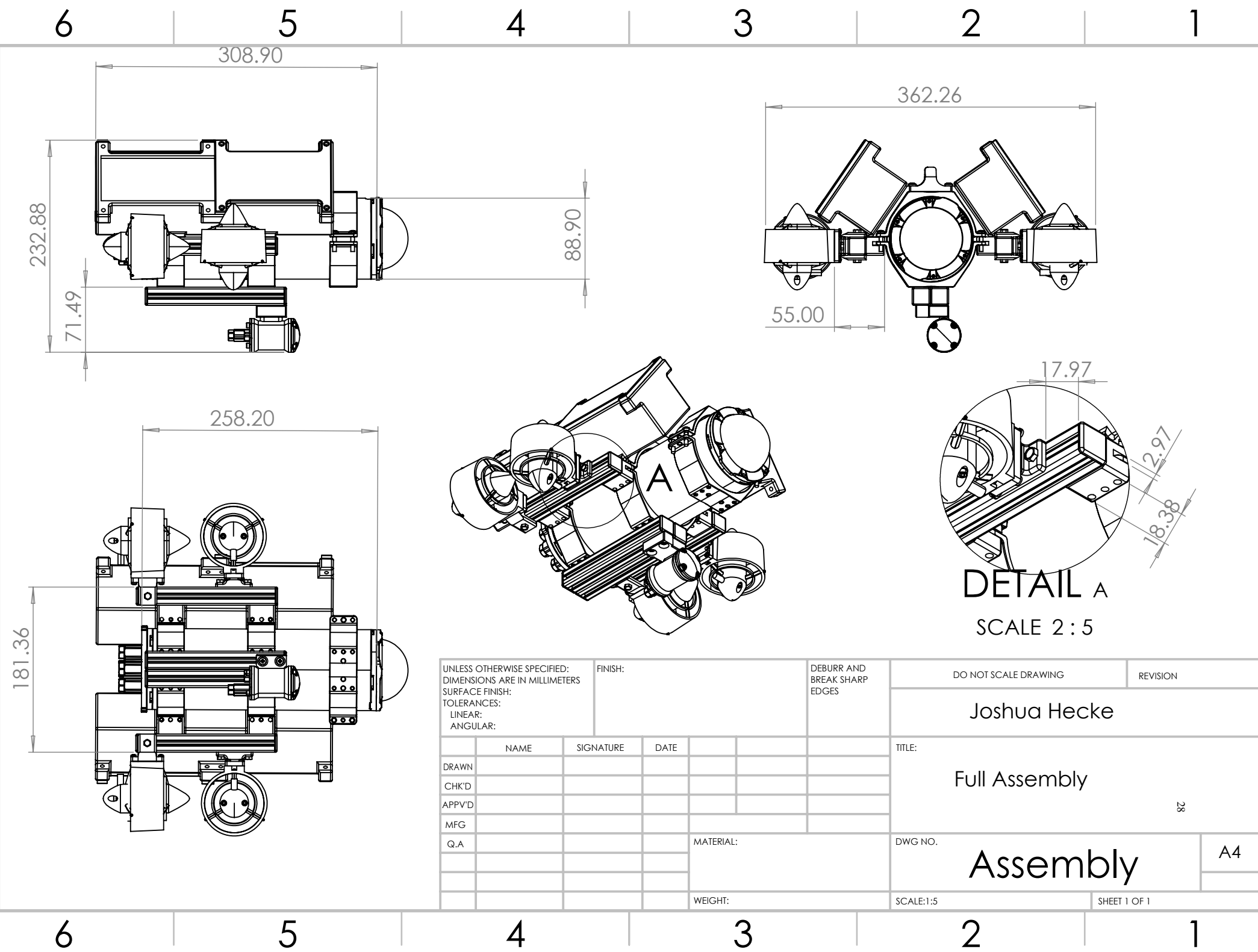


REVISED GANTT CHART

Project Name: SubbyROV	COMPANY NAME	QUT
Supervior: Tobias Fischer	DATE	16/2/2026







D

C

B

A

D

C

B

A

E. Appendix E: Risk Assessment

Customer Details:-

Owner:	Josh Hecke (joshua.hecke@connect.qut.edu.au)	Review:	Yes
Email:	joshua.hecke@connect.qut.edu.au	Endorsement:	Yes
Assessment Type:	Plant/Equipment	MAPS ID:	
Division/Faculty:	Faculty of Engineering	QUT Team:	School of Electrical Engineering & Robotics
School/Depart:	School of Electrical Engineering & Robotics	QUT SubTeam:	
Start Date:	8/25/2025	Est End Date:	6/25/2026
Title:	Design of small scale Remotely Operated Vehicle - Underwater Robot		
Description:	This is an risk assessment to support the design of a new small scale ROV on campus.		
	The primary risks relate to electrical safety during development, manual handling, operation, electrical safety including charging and packing and storage for travel.		
	Only Tobias Fischer, Joshua Hecke and Scarlet Raine will be able to operate this device during development and testing, due to its experimental nature. Other team members can operate this vehicle after contacting said operators and receiving a induction.		
	Main Risks- *5200mAh 14v LIPO Battery Use and Handling* *Operation and Handling around water* IF USED *Safe use and Storage of Dichtol Waterproofing* Ensure- S9 lab techs are aware of LIPO presence if battery is on campus for safe storage and charging.		
	In addition to this risk assessment, Approval of any commercial areas(i.e qut pool) needs to be organized with the appropriate body accordingly.		

Location Details:-

Campus	Building	Floor	Room	Additional Info
Off Campus				These are operated on-water in the field at approved locations(A key example would be QUT Pool or Personal Pool)
Gardens Point	S Block	Level 11	1145-Post-Graduate Lab.	Mobile Robots Lab Soldering Station

Hazard Details:-

Haz ID	Hazard	Cause of Harm	What could go wrong	L	C	Risk Score	Hide Items
79996	Slip-Trip-Fall	Slippery surfaces	Slipping getting in and out of boat, on boat ramps or on shore close to water.	Possible	Minor	Low	
		Control	Hierarchy: Administration Control: Assess conditions before moving around location. Must wear suitable footwear according to location (e.g. shoes, or wet booties). Only hop onto boat once fully secured. Do not run. Use multiple people to carry equipment on shore or loading boat.				
79997	Plant and Equipment	Operation	Rotating propellers - cuts, injuries, equipment in motion.	Unlikely	Minor	Low	
		Control	Hierarchy: Engineering Control: For the BlueRov, the remote operator interface controls the motors and touching any control will disable the system and put it into idle mode.				
		Control	Hierarchy: Isolation Control: On the BlueRov, the motors have shrouds minimizing the possibility of accidental contact with the propellers.				

79998	Manual Tasks	Posture - sustained/awkward/static	During deployment on shore/surface: - awkward posture (kneeling, squatting, balancing etc) - bending/twisting - duration of work - load handling - long standing/sitting - muscular force exerted - repetitive movement	Unlikely	Moderate	Medium	
		Control	Hierarchy: Other Control: Take regular breaks, Keep area tidy, Position body (and maneuver boat) to life appropriately Use the handles on the equipment				
79999	Electrical	<50 volts AC/<120 volts DC	Fire or damage to batteries due to charging or discharging	Possible	Minor	Low	
		Control	Hierarchy: Administration Control: After being approved/checked by lab tech, operators can now keep batteries on their person and conduct charging in designated charging areas in s11.				
		Control	Hierarchy: Administration Control: Follow the operating manual for the chargers and ASVs				
		Control	Hierarchy: Engineering Control: Only use official supplier and tagged chargers.				
		Control	Hierarchy: PPE Control: Ensure batteries are charged and transported in lipo safe bags				
		Control	Hierarchy: PPE Control: Fire extinguisher and/or fire blanket at working site.				
80000	Other	General assembly, building and installation of sensors and payloads.	- injury due to misuse of tools - pinch points	Possible	Minor	Low	
		Control	Hierarchy: Administration Control: Standard operating procedures General assembly of the main hull and location of pinch points are clearly marked. An online SOP for the ASVs is maintained (QUT wiki). Each participant is aware and has access to these manuals and the latest is attached to this risk assessment.				
		Control	Hierarchy: Administration Control: Training and education Ensure each participant is appropriately trained for the tools and equipment they are using and if required inducted into the work-zone.				
80001	Fieldwork	Other	Public hazards Unauthorized persons approach the vessel during operation such as swimmers or canoes. - knocking someone into the water - collision with a swimmer - collision with another vessel - public interfering with the vessel	Rare	Minor	Negligible	
		Control	Hierarchy: Administration Control: Standard Operating Procedures Follow the SOP and exception permit requirements for notifying the public and locations for operation.				
		Control	Hierarchy: Engineering Control: The overall mass is to be kept low so that there is minimal kinetic energy in any collision. Ensure all motors are shrouded and the shrouds are intact.				
		Control	Hierarchy: Isolation Control: Remote control devices All vessels have at least one level of wireless estop of wireless e-Stops. For the BlueRov2, the first is a hardwired heartbeat which must be maintained from the shore to allow power to the motors. The other is the remote control unit (e.g., RC controller or tablet) which can manually override the computer and when switched off will also trigger the e-Stop.				

80001		Control	Hierarchy: Isolation Control: The test location will be selected (moved within the site) to areas where there are no public visible or planned to be in the area. Conditions of applicable permits will be followed. If possible, on inland water storages, select sites that have no public access and seek approval from the land owners.				
80003	Fieldwork	Other	Working near water - drowning - slipping and falling - cuts from debris in the mud (e.g. broken glass, sharp rocks)	Unlikely	Major	Medium	
		Control	Hierarchy: Administration Control: Prepare and train for emergency response Details of the trained first-aiders made aware to the team on arrival at site as well as the location (in-vehicle) of the emergency contacts and closest hospital.				
		Control	Hierarchy: Administration Control: Standard operating procedures The captain on boarding the boats will advise the team and crew of the standard operating procedures for all aspects of working at sea. This includes location of life-jackets, procedure for man overboard. These procedures are documented on-board the vessels and placards are placed around the vessel. For people that cannot swim or are uncomfortable around the water, they are encouraged to wear life-jackets whenever working near the edge of the boat.				
		Control	Hierarchy: Administration Control: WAM-V SOP - Initially, the boat will be assembled on the waters edge following a visual inspection of the area and assessment of water depth. With a bow rope attached, the boat will be carried until the the stern is in the water (approximately 5 inches deep) and then pushed from the bow until completely in the water. Recovery will involve remote controlling the boat to the shore bow first, dragging the boat back onto shore (lifting the bow slightly) until the rear handles are at the shoreline and then lifted onto shore. This means that only shallow water sites are needed				
		Control	Hierarchy: PPE Control: Each participant will wear enclosed footwear to allow walking at the water edge without risk of cutting feet due to debris and with enough grip to minimise slipping.				
		Control	Hierarchy: PPE Control: Ensure suitable and fit for purpose life-jackets are available for all personal. These must be worn whilst on the boats when required by the skippers and vessels operating procedures.				
80004	Manual Tasks	Setting up equipment/furniture	Strains and sprains. Lifting equipment and transport cases for launching from shore.	Unlikely	Minor	Low	
		Control	Hierarchy: Administration Control: Assess weight and size of load. Use more people to lift if awkward, slippery or unable to lift. Use trolleys on campus.				
		Control	Hierarchy: Other Control: General assembly of the main hull and location of pinch points are detailed in the manufacturers manual (see attachment). Each participant is aware and has access to this manual.				
80005	Electrical	<50 volts AC/<120 volts DC	General hazard: Working with ultra low voltage DC (5-24Volts) Electrical equipment such as electric motors and sensors. - low voltage shock - burns - electrical fires	Possible	Minor	Low	
		Control	Hierarchy: Administration Control: Ensure all participants working with low-voltage electrical are trained in the use of the equipment and have the appropriate laboratory inductions if required to undertake the work.				
		Control	Hierarchy: Administration Control: Follow procedure for installation and removal of batteries. Ensure the equipment is turned off before installation or removal.				

80005		Control	Hierarchy: Administration Control: Keeping equipment and plant well maintained. Use only fit-for purpose equipment. Ensure all wiring is documented and appropriately labelled where necessary.				
		Control	Hierarchy: Isolation Control: Isolation switches and/or quick release plugs/connectors (such as Anderson or XT connectors) for isolating power to the components being worked on.				
		Control	Hierarchy: Other Control: Fire extinguisher and/or fire blanket at the work site. Also LiPo safe bags for transporting and charging LiPo batteries				
		Control	Hierarchy: PPE Control: Ensure batteries are charged and transported in lipo safe bags				
80006	Other	Loss of robot during missions	- Losing contact with the robot during missions - Leaking of water into robot	Unlikely	Minor	Low	
		Control	Hierarchy: Administration Control: Standard operating procedures The missions are planned such that the ROV will operate in within a defined range. The ROV are tethered and in loss of power the robot can be located and most likely retrieved using the tether. In the event of leaking of water into the ASV, the robot should have sufficient remaining buoyancy to remain at the surface In the event of a leaking robots, the batteries have been selected to minimize risk of fire or explosion and on recovery the robot will be isolated for up to 3 hours to ensure no fire risk before further handling.				
		Control	Hierarchy: Administration Control: The Operation of the ROV in navigable waters are covered under the AMSA Specific Exemption permit. This has requirements in the equipment required such as radios for alerting people in the vicinity, markings and the use of a ground control station (tablets and/or computers).				
		Control	Hierarchy: Other Control: The BlueRov have a heartbeat sent over the teather. The operational status and approximate position are always known within a few meters.				
80008	Electrical	Operation of electrical equipment	Burn, Shock, Breathing in solder fumes	Possible	Minor	Low	
		Control	Hierarchy: PPE Control: Ensure Glasses and extraction fan are used when soldering.				
		Control	Hierarchy: PPE Control: When handling parts and chemicals ensure gloves are worn and safety gear including masks are worn.				
80178	Chemicals	Handling and use	*This chemical may not be used* Use of non-hazardous dichlor for waterproofing 3d parts. Standard handling risks include inhalation, skin contact, eye contact	Possible	Minor	Low	
		Control	Hierarchy: Administration Control: The chemical will be stored in s9 store when not in use and be lent out when used in s11. To be returned at end of use				
		Control	Hierarchy: Isolation Control: Remove risk by storing in carefully closed container upright with lid on tight				

80178		Control	Hierarchy: Other Control: General information In all cases of doubt, or when symptoms persist, seek medical advice. If unconscious but breathing normally, place in recovery position and seek medical advice. Following inhalation Remove casualty to fresh air and keep warm and at rest. In case of irregular breathing or respiratory arrest provide artificial respiration. Following skin contact Remove contaminated, saturated clothing immediately. After contact with skin, wash immediately with plenty of water and soap. Do not use solvents or thinners. Wash contaminated clothing before reuse. After eye contact Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Seek medical advice immediately. Following ingestion If swallowed, rinse mouth with water (only if the person is conscious). Seek medical advice immediately. Keep victim calm. Do NOT induce vomiting. Self-protection of the first aider First aider: Pay attention to self-protection! Most important symptoms and effects, both acute and delayed Symptoms In all cases of doubt, or when symptoms persist, seek medical advice. Indication of any immediate medical attention and special treatment needed First Aid, decontamination, treatment of symptoms.				
		Control	Hierarchy: PPE Control: gloves, eye protection, respiratory protection				
80320	Electrical	Design/modification	Water could enter electrical storage containers causing damage and potential shock. This is due to waterproof seals failing or not being sufficient	Possible	Moderate	Medium	
		Control	Hierarchy: Elimination Control: Try to use certified/standard waterproofed containers (BlueRov product) where possible and only use custom made containers where necessary				
		Control	Hierarchy: Engineering Control: Ensure all waterproof containers are tested multiple times at the required depth before being used to store components.				
83411	Electrical	Operation of electrical equipment	Lipo Battery 5200mAh 14V -Overcharging- can cause damage and fires -Storage- Storing LiPo batteries in inappropriate temperatures or locations can affect their performance and safety -Thermal Runaway-A dangerous process where a battery overheats and cannot cool down, leading to fire, explosion, and toxic fumes -Physical Damage- Punctures, fractures, or tears in the battery casing can lead to internal short circuits, increasing the risk of fire	Possible	Moderate	Medium	
		Control	Hierarchy: Administration Control: Ensure days notice before any use, s9 techs will prepare accordingly.				
		Control	Hierarchy: Administration Control: For Handling, Lab Techs in s9 control the charging and storage of LIPOs. This is non-negotiable as the methods already in place for safe storage, charging and testing.				
		Control	Hierarchy: Elimination Control: Check resistance between battery input terminals and ensure no shorts or low resistance before use to ensure max output is not placed on system.				

83411		Control	Hierarchy: Engineering Control: Before the battery is in operation ensure use in the system doesn't heat up the battery to a safety critical range. This is done through safe testing and calculations to ensure constant high draw causes heat				
		Control	Hierarchy: Engineering Control: Don't run the battery down below 20% capacity.				
		Control	Hierarchy: Isolation Control: Ensure when the battery is in use and/or exposed that there is a safe distance/exclusion zone. Especially for initial testing and research.				
		Control	Hierarchy: Isolation Control: The LIPO will remain inside a LIPO Bag during transport at all times. Ensure it is collected like this.				

Participant Details:-

Participant Name
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Reviewer Name	Review Status
Tobias Fischer (tobias.fischer@qut.edu.au)	Pending
Steven Bulmer (steven.bulmer@qut.edu.au)	Pending

Approver Details:-

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Dhammika Jayalath (dhammika.jayalath@qut.edu.au)

Approval Details:-

ID	Approval	Approver Name	Approval Status
21769	Risk Assessment Endorsment required for Risk Assessment#18339	Tobias Fischer (tobias.fischer@qut.edu.au)	Approved
22118	Risk Assessment Approval required for Assessment #18339	Firuz Zare (f.zare@qut.edu.au)	Cancelled
22165	Risk Assessment Endorsment required for Risk Assessment#18339	Tobias Fischer (tobias.fischer@qut.edu.au)	Pending

Note Details:-

Subject	Note	Category	Created Date	Created By
Reviewer Note	Hi Josh, Overall well done and comprehensive RA! - I've added Scarlett Raine as a participant as she's leading a lot of our marine robotics work, please update description accordingly. - Other team members need to contact operators + get induction - Remove everything boat related from the RA (loss of robot, ...), this is out of scope for this RA. If boat operation will happen in the future, we'll need to submit a separate RA. - Need to add water leakage as hazard + appropriate controls; describe tank etc. - Need to provide detailed information on where chemicals will be stored (Krishna to advise) - Need to provide detailed information on where batteries will be stored (Krishna to advise) - Change "low voltage" to "ultra low voltage" to be consistent with regulation Let me know if you have any questions. Best, Tobi	Customer Notes	9/19/2025 10:29:17 AM	Tobias Fischer (tobias.fischer@qut.edu.au)

Reviewer Note	Happy to accept this for now but maybe some additions would help further readings. Make it clear that the batteries used are not LIPO. I was sent a link to a Li-Ion battery which is a much safer chemistry. I would get Shaun and or S9 involved asap to get him in the loop for chemical purchase and storage. I think ultimately the users will be the ones requesting the chemicals for purchase and then involve S9 for the storage.	Customer Notes	9/19/2025 10:29:17 AM	Steven Bulmer (steven.bulmer@qut.edu.au)
Endorser Note	Hi Josh, I am happy to endorse this as soon as 1) Scarlett has been added to Project Description, and 2) You consider Steve's comments about chemicals/battery descriptions. Best, Tobi	Customer Notes	9/19/2025 10:29:17 AM	Tobias Fischer (tobias.fischer@qut.edu.au)